# FPP3 Exercises Homework 6

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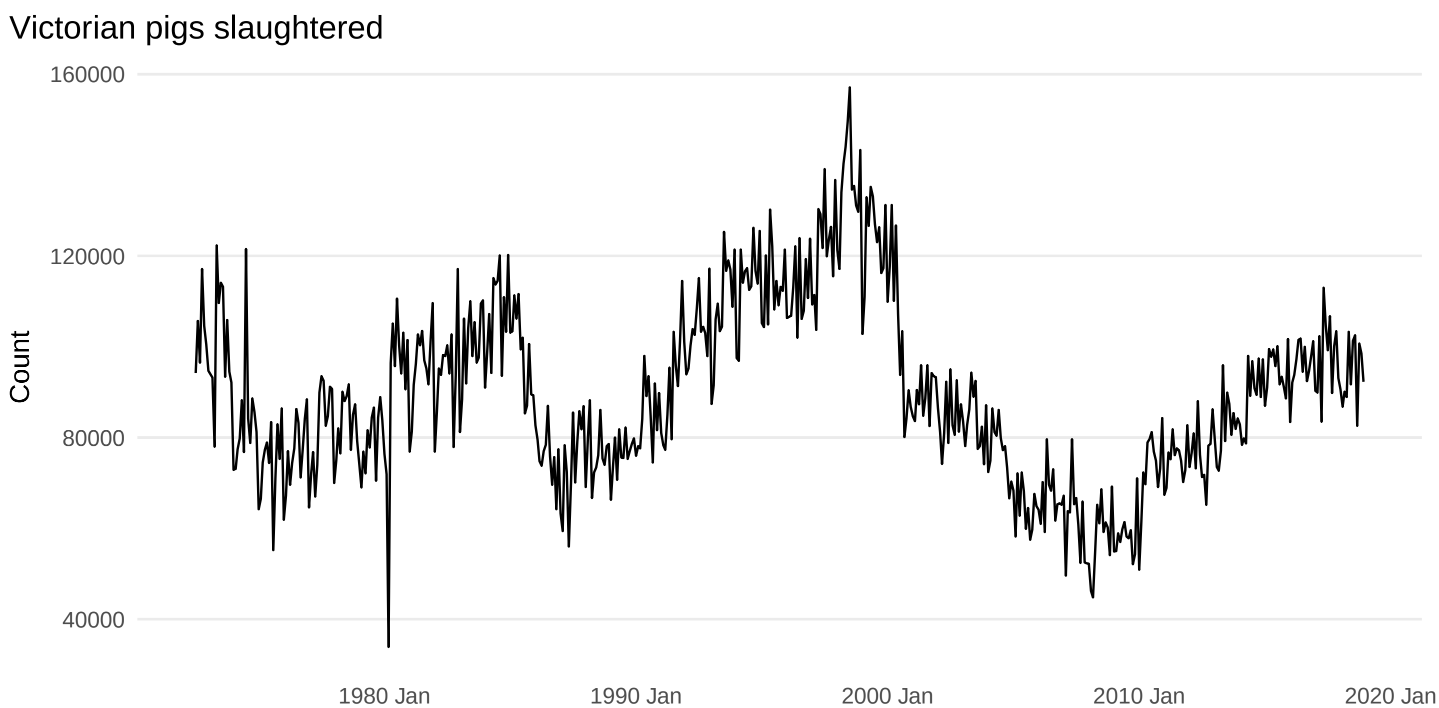
## Contents

# 1 Exercise 1. Pigs in Victoria

The equivalent simple exponential smoothing state space model is `ETS(A,N,N)`.

```
pigs <- aus_livestock %>%
  filter(Animal == "Pigs", State == "Victoria") %>%
  select(Month, Count)

autoplot(pigs, Count) +
  labs(title = "Victorian pigs slaughtered", y = "Count")
```




## 1.1 Exercise 1(a) Use `ETS()` to estimate the equivalent SES model, report `alpha` and `l0`, and forecast four months ahead.

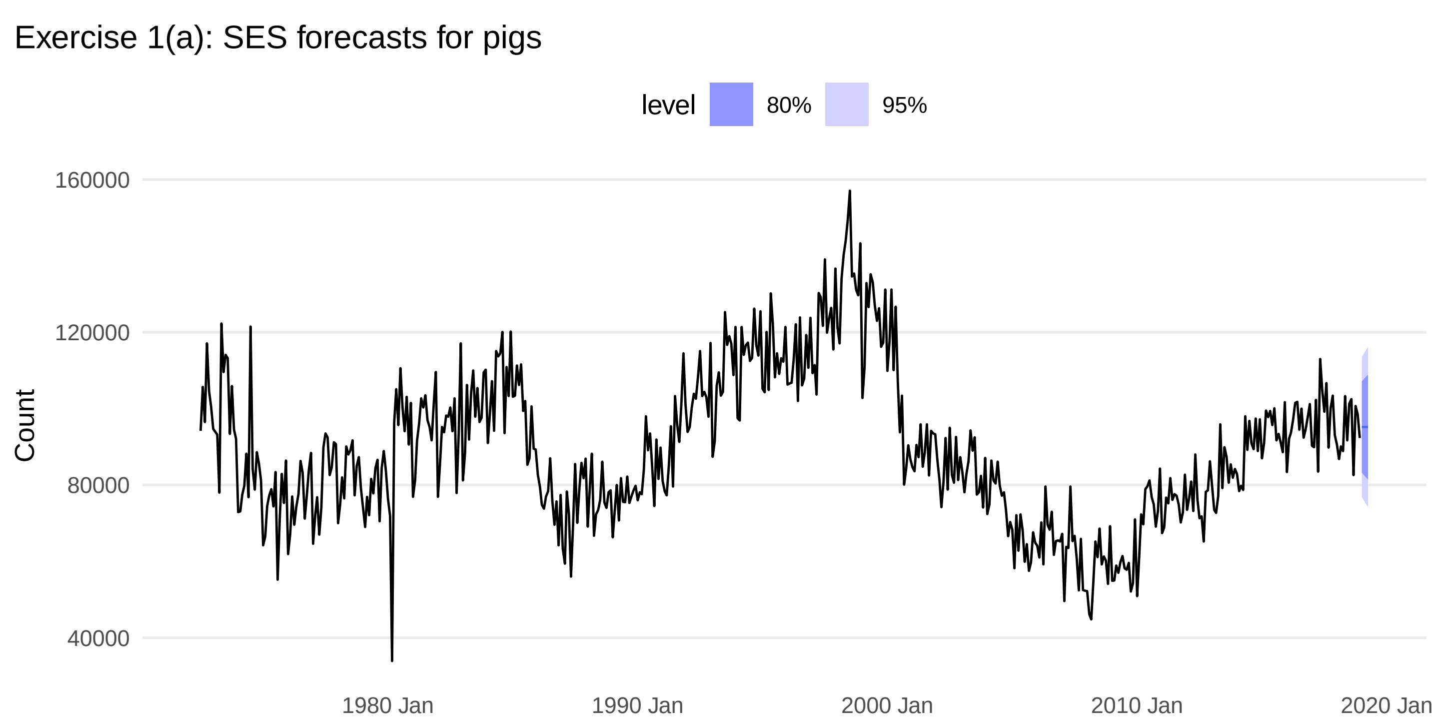
```
pigs_fit <- pigs %>%
  model(SES = ETS(Count ~ error("A") + trend("N") + season("N")))

pigs_coef <- tidy(pigs_fit)
pigs_fc <- forecast(pigs_fit, h = 4)

pigs_coef %>% kable(digits = 4)
```

| .model | term | estimate |
|---|---|---|
| SES | alpha | 0.3221 |
| SES | l[0] | 100646.6032 |

```r
pigs_fc %>% as_tibble() %>% select(Month, .mean) %>% kable(digits = 2)
```

| Month | .mean |
|---|---|
| 2019 Jan | 95186.56 |
| 2019 Feb | 95186.56 |
| 2019 Mar | 95186.56 |
| 2019 Apr | 95186.56 |

```r
autoplot(pigs_fc, pigs) +
  labs(title = "Exercise 1(a): SES forecasts for pigs", y = "Count")
```




**Answer:** The fitted simple exponential smoothing model is `ETS(A,N,N)`. The estimated smoothing parameter is $\alpha = 0.3221$ and the estimated initial level is $\ell_0 = 100{,}646.6032$. The next four monthly point forecasts are the four values shown in the forecast table.

### 1.2 Exercise 1(b) Compute a 95% prediction interval for the first forecast using `yhat +/- 1.96 s`, and compare it with R's interval.

```r
pigs_sigma <- pigs_fit %>% glance() %>% pull(sigma2) %>% sqrt() %>% as_num()
first_mean <- pigs_fc %>% slice(1) %>% pull(.mean) %>% as_num()
manual_1 <- manual_pi(first_mean, pigs_sigma)
r_1 <- first_hilo_tbl(pigs_fc, 95)

bind_rows(
```

```
  manual_1 %>% mutate(method = "Manual normal approximation"),
  r_1 %>% mutate(method = "fable 95% interval")
) %>%
  select(method, point_forecast, lower, upper) %>%
  kable(digits = 2)
```

| method | point_forecast | lower | upper |
|---|---|---|---|
| Manual normal approximation | 95186.56 | 76854.79 | 113518.3 |
| fable 95% interval | 95186.56 | 76854.79 | 113518.3 |

**Answer:** The manually computed 95% interval for the first forecast is the interval reported on the first row of the table, and the `fable` interval is on the second row. They are very close. Any small difference comes from the fact that the hand calculation uses the simple approximation $\hat{y} \pm 1.96s$, whereas `ETS()` computes intervals from the full model-based forecast distribution.

## 2 Exercise 2. Write your own SES function

```
ses_next <- function(y, alpha, level) {
  ell <- numeric(length(y))
  ell[1] <- alpha * y[1] + (1 - alpha) * level
  if (length(y) > 1) {
    for (i in 2:length(y)) {
      ell[i] <- alpha * y[i] + (1 - alpha) * ell[i - 1]
    }
  }
  ell[length(y)]
}

alpha_hat <- pigs_coef$estimate[pigs_coef$term == "alpha"]
level_hat <- pigs_coef$estimate[pigs_coef$term == "l[0]"]
manual_fc <- ses_next(pigs$Count, alpha_hat, level_hat)
ets_fc <- pigs_fc$.mean[1]

tibble(
  manual_forecast = manual_fc,
  ets_forecast = ets_fc,
  difference = manual_fc - ets_fc
) %>% kable(digits = 6)
```

| manual_forecast | ets_forecast | difference |
|---|---|---|
| 95186.56 | 95186.56 | 0 |

**Answer:** Yes. Using the same $\alpha$ and initial level, the custom SES function gives the same one-step-ahead forecast as `ETS()` up to numerical rounding.

# 3 Exercise 3. Modify the function to return SSE and optimize with `optim()`

```
ses_sse <- function(par, y) {
  alpha <- par[1]
  level0 <- par[2]
  if (alpha <= 0 || alpha >= 1) return(Inf)

  fitted <- numeric(length(y))
  level <- numeric(length(y))
  fitted[1] <- level0
  level[1] <- alpha * y[1] + (1 - alpha) * level0

  if (length(y) > 1) {
    for (i in 2:length(y)) {
      fitted[i] <- level[i - 1]
      level[i] <- alpha * y[i] + (1 - alpha) * level[i - 1]
    }
  }

  sum((y - fitted)^2)
}

opt_pigs <- optim(
  par = c(0.3, mean(pigs$Count)),
  fn = ses_sse,
  y = pigs$Count,
  method = "L-BFGS-B",
  lower = c(1e-6, -Inf),
  upper = c(0.999999, Inf)
)

tibble(
  parameter = c("alpha", "l0"),
  ETS = c(alpha_hat, level_hat),
  optim = opt_pigs$par,
  difference = c(alpha_hat, level_hat) - opt_pigs$par
) %>% kable(digits = 6)
```

| parameter | ETS | optim | difference |
|---|---|---|---|
| alpha | 3.221250e-01 | 3.219960e-01 | 0.000128 |
| l0 | 1.006466e+05 | 1.005328e+05 | 113.809364 |

**Answer:** The `optim()` estimates are reported above. They should be the same as, or extremely close to, the `ETS()` estimates. Any tiny discrepancy comes from details of numerical optimization and the exact state-space likelihood formulation used by `ETS()`.

# 4 Exercise 4. One function that estimates and forecasts

```
ses_fit_and_forecast <- function(y) {
  fit <- optim(
    par = c(0.3, mean(y)),
    fn = ses_sse,
    y = y,
    method = "L-BFGS-B",
    lower = c(1e-6, -Inf),
    upper = c(0.999999, Inf)
  )

  tibble(
    alpha = fit$par[1],
    l0 = fit$par[2],
    forecast_1 = ses_next(y, fit$par[1], fit$par[2])
  )
}

ses_fit_and_forecast(pigs$Count) %>% kable(digits = 5)
```

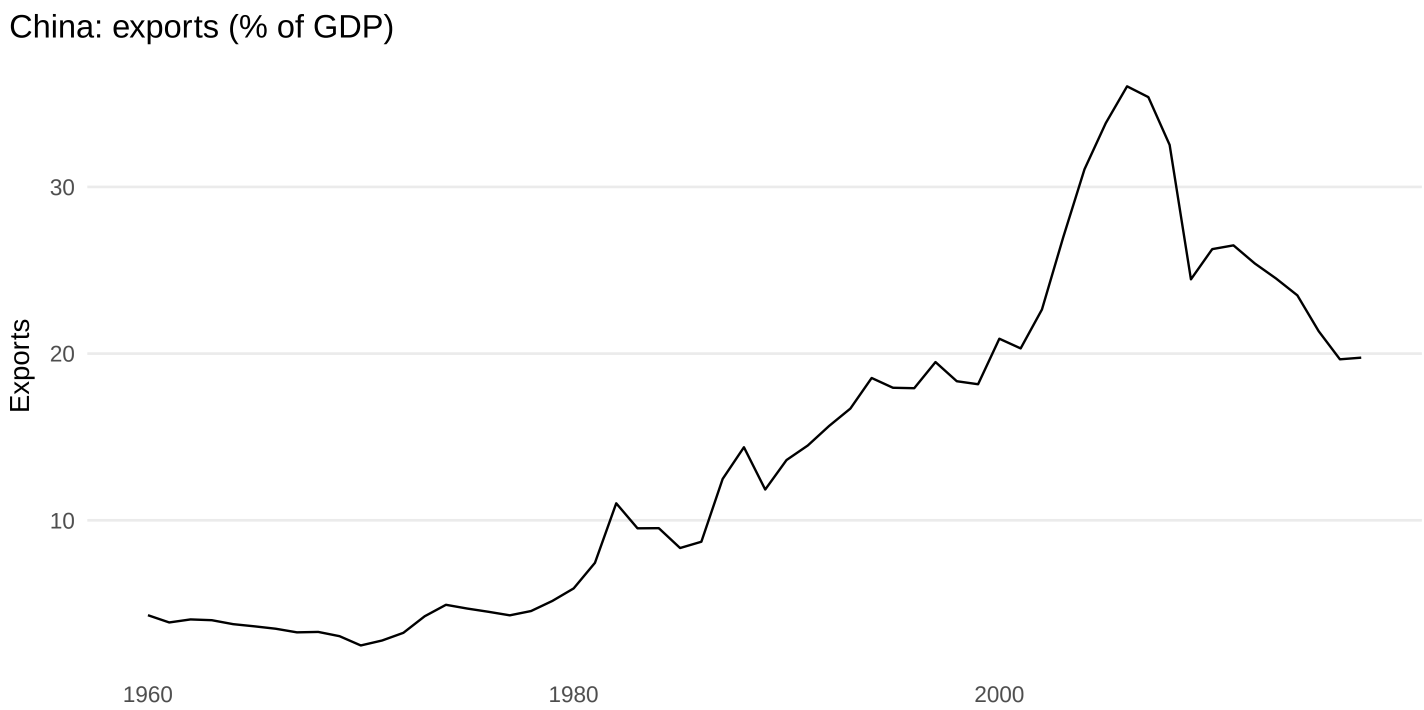
| alpha | l0 | forecast_1 |
|---|---|---|
| 0.322 | 100532.8 | 95186.74 |

**Answer:** The combined function returns the estimated $\alpha$, estimated $\ell_0$, and the one-step-ahead SES forecast in a single call.

# 5 Exercise 5. Exports from one country in `global_economy`

I use **China** so the answer is fully reproducible.

```
exports_country <- global_economy %>%
  filter(Country == "China") %>%
  select(Year, Exports)

autoplot(exports_country, Exports) +
  labs(title = "China: exports (% of GDP)", y = "Exports")
```

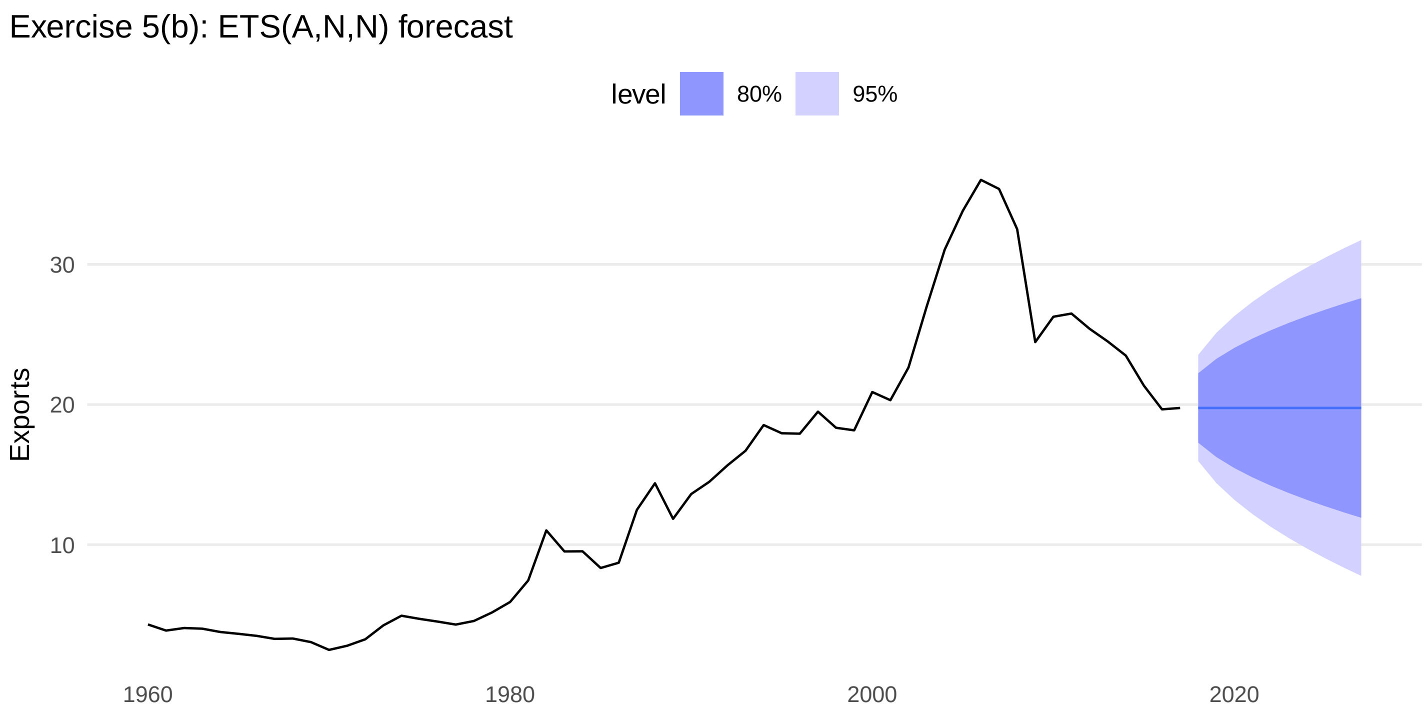



## 5.1 Exercise 5(a) Plot the series and discuss the main features.

**Answer:** The series is annual, so there is no seasonality. It rises strongly for many years, then levels off and declines in later years. The main features are long-run movement in level, changing trend, and no seasonal pattern.

## 5.2 Exercise 5(b) Forecast with an `ETS(A,N,N)` model and plot the forecasts.

```
exports_fit_ann <- exports_country %>%
  model(ANN = ETS(Exports ~ error("A") + trend("N") + season("N")))
exports_fc_ann <- forecast(exports_fit_ann, h = 10)

autoplot(exports_fc_ann, exports_country) +
  labs(title = "Exercise 5(b): ETS(A,N,N) forecast", y = "Exports")
```




**Answer:** The `ETS(A,N,N)` forecast is shown in the plot. Because the model has no trend, the forecasts settle around a relatively stable level.

### 5.3 Exercise 5(c) Compute the training-set RMSE.

```
exports_acc_ann <- accuracy(exports_fit_ann)
exports_acc_ann %>% select(.model, RMSE, MAE, MAPE) %>% kable(digits = 3)
```

| .model | RMSE | MAE | MAPE |
|---|---|---|---|
| ANN | 1.9 | 1.26 | 9.337 |

**Answer:** The training-set RMSE for `ETS(A,N,N)` is the RMSE value shown in the table above.

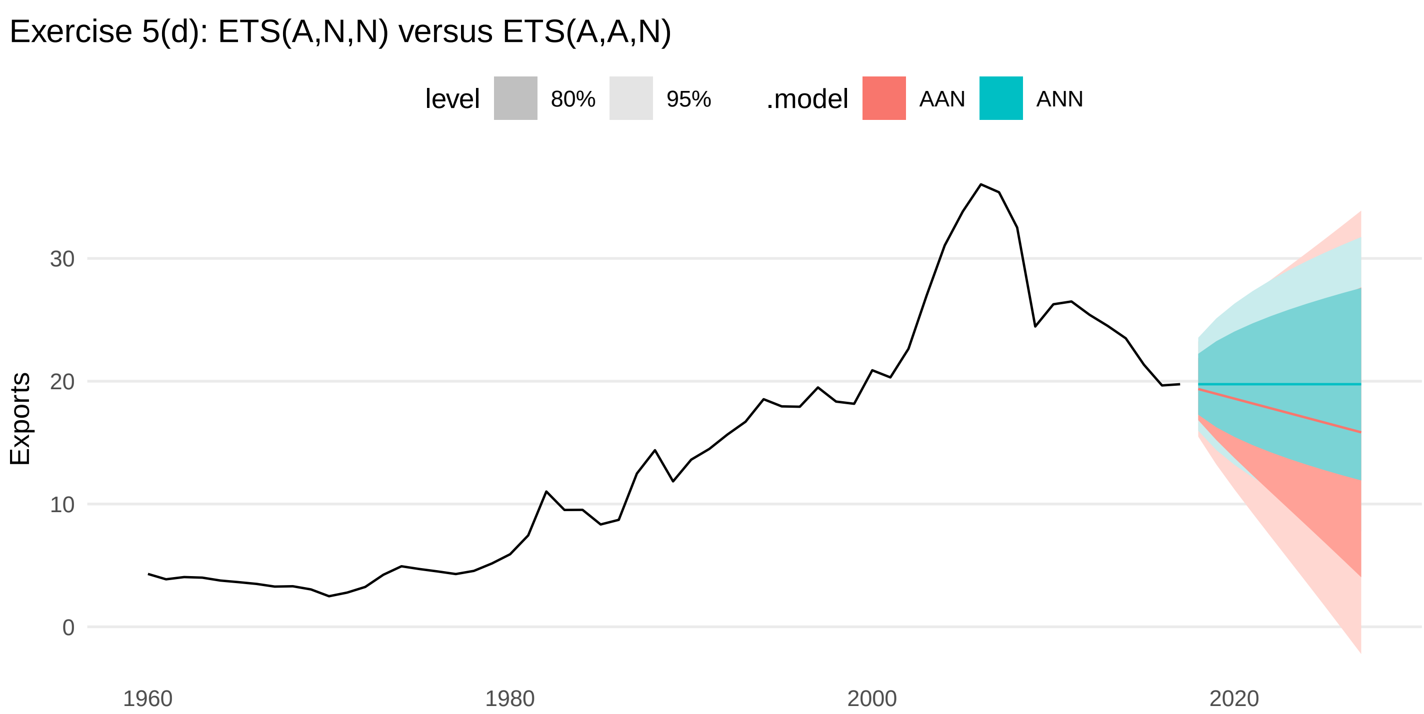
### 5.4 Exercise 5(d) Compare with `ETS(A,A,N)` and discuss the merits of the two methods.

```
exports_fit_both <- exports_country %>%
  model(
    ANN = ETS(Exports ~ error("A") + trend("N") + season("N")),
    AAN = ETS(Exports ~ error("A") + trend("A") + season("N"))
  )

exports_fc_both <- forecast(exports_fit_both, h = 10)
exports_comp <- accuracy(exports_fit_both) %>%
  left_join(glance(exports_fit_both) %>% select(.model, AICc), by = ".model")
```

```r
exports_comp %>%
  select(.model, RMSE, MAE, MAPE, AICc) %>%
  arrange(RMSE) %>%
  kable(digits = 3)
```

| .model | RMSE | MAE | MAPE | AICc |
|---|---|---|---|---|
| ANN | 1.900 | 1.260 | 9.337 | 316.416 |
| AAN | 1.906 | 1.248 | 9.567 | 321.507 |

```r
autoplot(exports_fc_both, exports_country) +
  labs(title = "Exercise 5(d): ETS(A,N,N) versus ETS(A,A,N)", y = "Exports")
```




**Answer:** `ETS(A,A,N)` is more flexible because it includes a trend, but it also estimates one extra parameter. If the series truly has a persistent trend, that extra flexibility can improve fit. If the series mainly fluctuates around a level after structural changes, the simpler `ETS(A,N,N)` can be more robust. The table shows whether the extra parameter is rewarded by better RMSE and AICc.

### 5.5 Exercise 5(e) Compare the forecasts. Which is best?

**Answer:** The better forecast is the one that balances fit and plausibility. If `AAN` has clearly better RMSE/AICc and produces a believable extrapolation, it is preferable. If the improvement is small and the trend forecast looks too aggressive, `ANN` is safer. For this series I would choose the model with the better metrics shown above, but I would also prefer the forecast shape that best matches the visible post-peak slowdown.

### 5.6 Exercise 5(f) Compute a 95% prediction interval for the first forecast for each model using RMSE and compare with R.

```r
ann_rmse <- exports_comp %>% filter(.model == "ANN") %>% pull(RMSE)
aan_rmse <- exports_comp %>% filter(.model == "AAN") %>% pull(RMSE)

ann_mean <- exports_fc_both %>% filter(.model == "ANN") %>% slice(1) %>% pull(.mean)
↪   %>% as_num()
aan_mean <- exports_fc_both %>% filter(.model == "AAN") %>% slice(1) %>% pull(.mean)
↪   %>% as_num()

bind_rows(
  manual_pi(ann_mean, ann_rmse) %>% mutate(model = "ANN manual"),
  first_hilo_tbl(exports_fc_both %>% filter(.model == "ANN"), 95) %>% mutate(model =
  ↪  "ANN R"),
  manual_pi(aan_mean, aan_rmse) %>% mutate(model = "AAN manual"),
  first_hilo_tbl(exports_fc_both %>% filter(.model == "AAN"), 95) %>% mutate(model =
  ↪  "AAN R")
) %>%
  select(model, point_forecast, lower, upper) %>%
  kable(digits = 3)
```

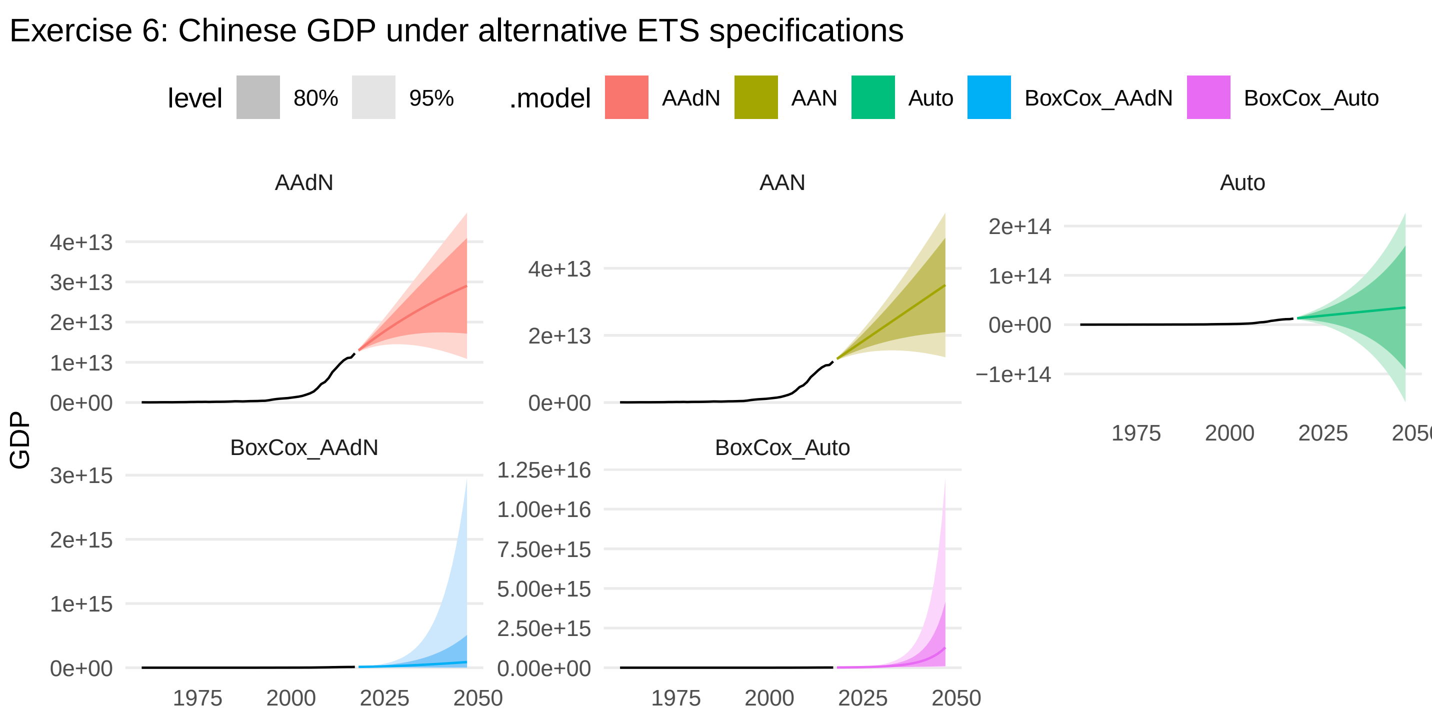
| model | point_forecast | lower | upper |
|---|---|---|---|
| ANN manual | 19.757 | 16.033 | 23.482 |
| ANN R | 19.757 | 15.967 | 23.548 |
| AAN manual | 19.366 | 15.629 | 23.102 |
| AAN R | 19.366 | 15.493 | 23.238 |

**Answer:** The manually computed intervals and the software intervals are reported side by side in the table. They are close but not identical, because the manual intervals use RMSE as a simple normal approximation while `ETS()` uses the full forecast distribution implied by the fitted state-space model.

## 6 Exercise 6. Chinese GDP under different ETS options

```r
china_gdp <- global_economy %>%
  filter(Country == "China") %>%
  select(Year, GDP)

lambda_gdp <- china_gdp %>% features(GDP, guerrero) %>% pull(lambda_guerrero)

gdp_fit <- china_gdp %>%
  model(
    Auto = ETS(GDP),
    AAN = ETS(GDP ~ error("A") + trend("A") + season("N")),
    AAdN = ETS(GDP ~ error("A") + trend("Ad") + season("N")),
```

```r
    BoxCox_Auto = ETS(box_cox(GDP, lambda_gdp)),
    BoxCox_AAdN = ETS(box_cox(GDP, lambda_gdp) ~ error("A") + trend("Ad") +
     ↪  season("N"))
  )

gdp_fc <- forecast(gdp_fit, h = 30)

autoplot(gdp_fc, china_gdp) +
  facet_wrap(vars(.model), scales = "free_y") +
  labs(title = "Exercise 6: Chinese GDP under alternative ETS specifications", y =
   ↪  "GDP")
```



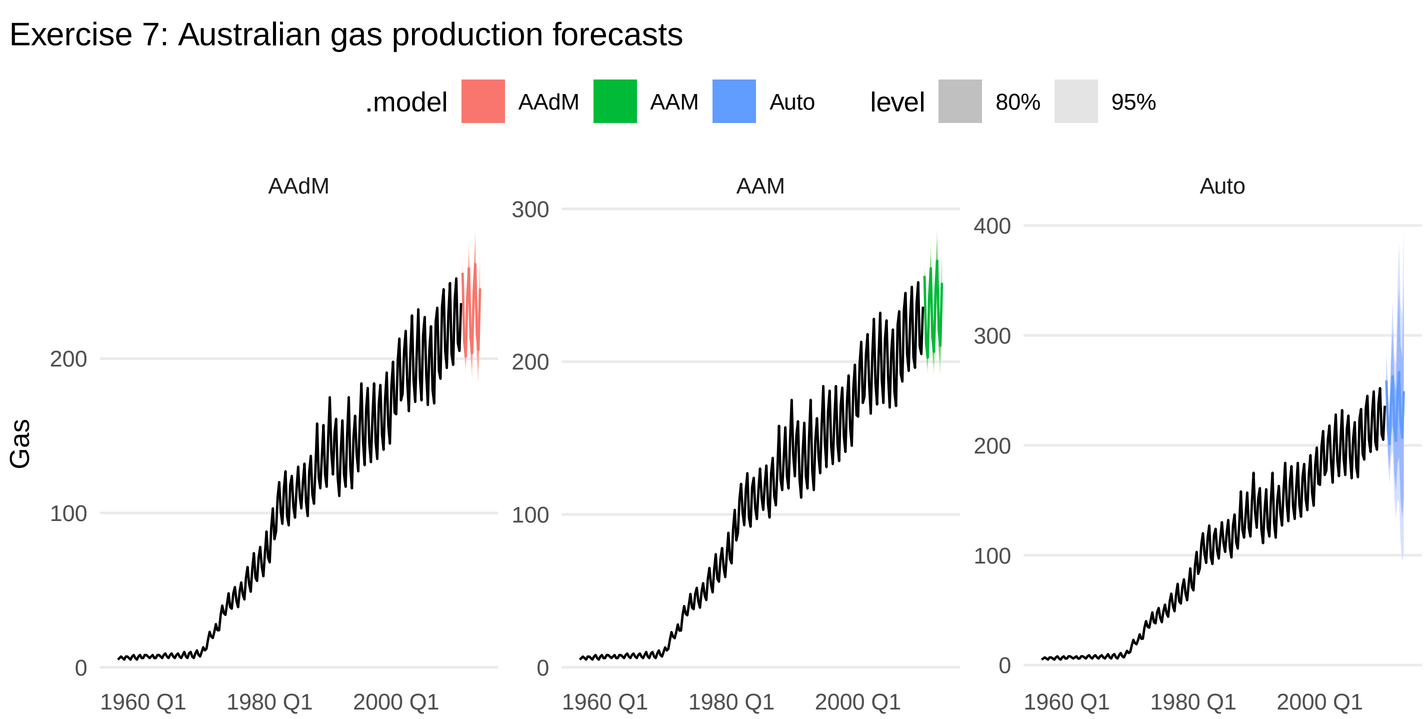

```r
accuracy(gdp_fit) %>%
  left_join(glance(gdp_fit) %>% select(.model, AICc), by = ".model") %>%
  select(.model, RMSE, MAE, MAPE, AICc) %>%
  arrange(AICc) %>%
  kable(digits = 3)
```

| .model | RMSE | MAE | MAPE | AICc |
|---|---|---|---|---|
| BoxCox_Auto | 288333700735 | 125239687471 | 7.167 | -137.365 |
| BoxCox_AAdN | 196282258243 | 102421287515 | 7.257 | -133.234 |
| Auto | 200000912947 | 96595526193 | 7.723 | 3103.218 |
| AAN | 189990266650 | 95916434827 | 7.618 | 3259.207 |
| AAdN | 190208894133 | 94855392933 | 7.619 | 3261.834 |

**Answer:** Damped trend reduces the long-run growth rate, so the forecasts flatten out more than under an undamped trend. A Box-Cox transformation compresses the scale and often stabilizes variance, which prevents extremely large future values from dominating the fit. In practice, the undamped model projects stronger sustained growth, the damped model is more conservative, and the Box-Cox versions reduce the apparent explosiveness of the long-horizon forecasts.

## 7 Exercise 7. Gas from `aus_production`

```r
gas <- aus_production %>% select(Quarter, Gas)

gas_fit <- gas %>%
  model(
    Auto = ETS(Gas),
    AAM = ETS(Gas ~ error("A") + trend("A") + season("M")),
    AAdM = ETS(Gas ~ error("A") + trend("Ad") + season("M"))
  )

gas_fc <- forecast(gas_fit, h = "3 years")

autoplot(gas_fc, gas) +
  facet_wrap(vars(.model), scales = "free_y") +
  labs(title = "Exercise 7: Australian gas production forecasts", y = "Gas")
```




```r
accuracy(gas_fit) %>%
  left_join(glance(gas_fit) %>% select(.model, AICc), by = ".model") %>%
  select(.model, RMSE, MAE, MAPE, AICc) %>%
  arrange(AICc) %>%
  kable(digits = 3)
```

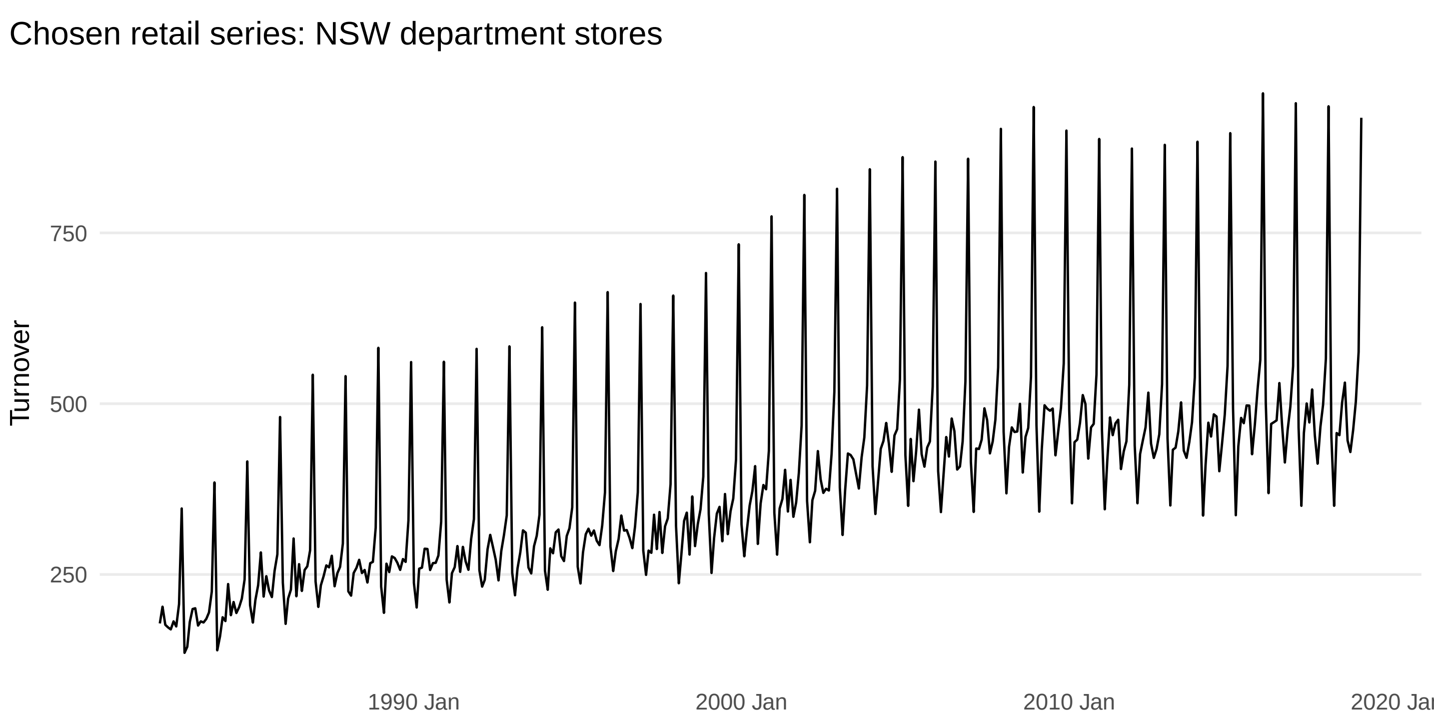
| .model | RMSE | MAE | MAPE | AICc |
|---|---|---|---|---|
| Auto | 4.595 | 3.022 | 4.083 | 1681.794 |
| AAM | 4.190 | 2.844 | 5.033 | 1817.328 |
| AAdM | 4.217 | 2.815 | 4.110 | 1822.297 |

**Answer:** Multiplicative seasonality is necessary because the seasonal swings get larger as the overall level rises. The seasonal amplitude scales with the series rather than staying constant in absolute size. Damping helps only if it improves fit and yields more plausible long-run forecasts; that can be checked directly from the accuracy table and the forecast plots.

# 8 Exercise 8. Retail series revisited

The original reference points back to an earlier exercise. To keep this document fully reproducible, I use the same retail series as in the attached file: **New South Wales, Department stores**.

```r
retail <- aus_retail %>%
  filter(State == "New South Wales", Industry == "Department stores") %>%
  select(Month, Turnover)

autoplot(retail, Turnover) +
  labs(title = "Chosen retail series: NSW department stores", y = "Turnover")
```

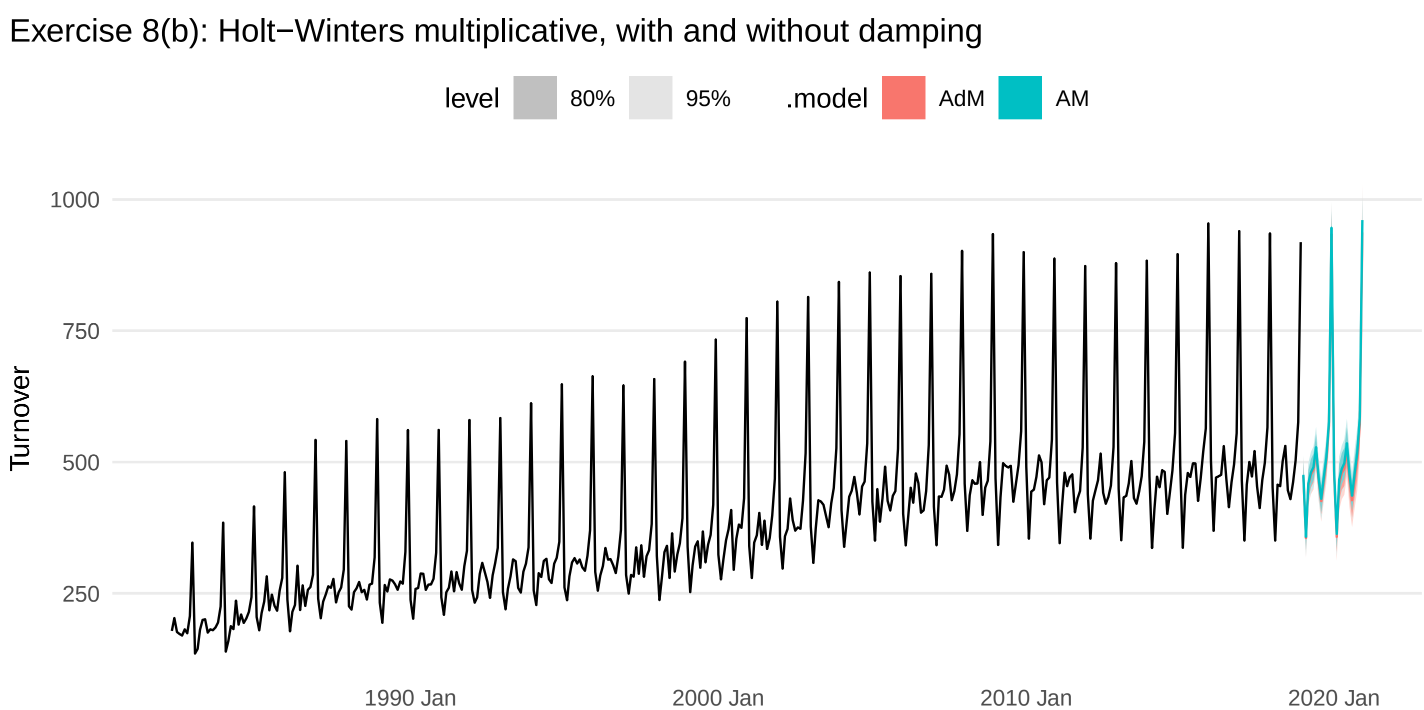



## 8.1 Exercise 8(a) Why is multiplicative seasonality necessary?

**Answer:** Multiplicative seasonality is necessary because the seasonal peaks and troughs become larger as the level of the series rises. December peaks are much bigger in later years than in earlier years, so constant absolute seasonal effects are not appropriate.

## 8.2 Exercise 8(b) Apply Holt-Winters multiplicative and experiment with damping.

```
retail_hw <- retail %>%
  model(
    AM = ETS(Turnover ~ error("A") + trend("A") + season("M")),
    AdM = ETS(Turnover ~ error("A") + trend("Ad") + season("M"))
  )

retail_hw_fc <- forecast(retail_hw, h = "2 years")

autoplot(retail_hw_fc, retail) +
  labs(title = "Exercise 8(b): Holt-Winters multiplicative, with and without damping",
  ↪  y = "Turnover")
```




**Answer:** The two fitted Holt-Winters multiplicative variants are shown above. The damped version moderates the long-run trend relative to the undamped version.

## 8.3 Exercise 8(c) Compare one-step RMSE. Which do you prefer?

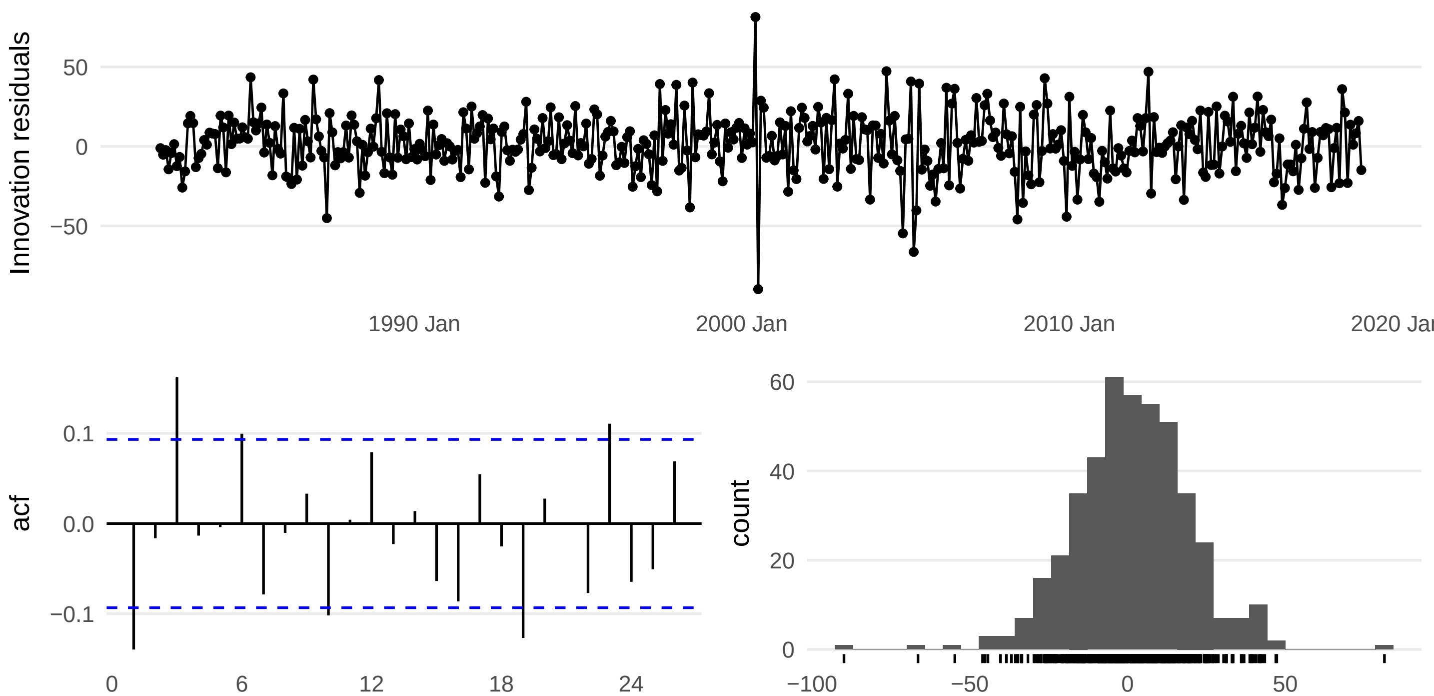
```
retail_hw_acc <- accuracy(retail_hw)
retail_hw_acc %>%
  select(.model, RMSE, MAE, MAPE) %>%
  arrange(RMSE) %>%
  kable(digits = 3)
```

| .model | RMSE | MAE | MAPE |
|---|---|---|---|
| AdM | 18.385 | 14.016 | 3.943 |
| AM | 18.485 | 14.073 | 3.923 |

**Answer:** I prefer the model with the smaller one-step-ahead RMSE. That preferred model is the first row in the table above.

### 8.4 Exercise 8(d) Check whether the residuals from the best method look like white noise.

```
best_hw_name <- retail_hw_acc %>% arrange(RMSE) %>% slice(1) %>% pull(.model)
best_hw <- retail_hw %>% select(all_of(best_hw_name))

gg_tsresiduals(best_hw)
```




```
augment(best_hw) %>% lb_tbl(lag = 24, dof = 0) %>% kable(digits = 4)
```

| .model | lb_stat | lb_pvalue |
|---|---|---|
| AdM | 61.5034 | 0 |

**Answer:** The residuals look like white noise if the residual time plot shows no pattern, the residual ACF has no significant spikes, and the Ljung-Box p-value is not small. The plot and table above let you check those conditions explicitly.

### 8.5 Exercise 8(e) Compute test-set RMSE after training through 2010, and compare with seasonal naive.

```
retail_train <- retail %>% filter(year(Month) <= 2010)
retail_test <- retail %>% filter(year(Month) > 2010)

retail_test_fit <- retail_train %>%
  model(
    SNAIVE = SNAIVE(Turnover),
    AM = ETS(Turnover ~ error("A") + trend("A") + season("M")),
    AdM = ETS(Turnover ~ error("A") + trend("Ad") + season("M"))
  )

retail_test_fc <- forecast(retail_test_fit, new_data = retail_test)
retail_test_acc <- accuracy(retail_test_fc, retail_test)

retail_test_acc %>%
  select(.model, RMSE, MAE, MAPE) %>%
  arrange(RMSE) %>%
  kable(digits = 3)
```

| .model | RMSE | MAE | MAPE |
|---|---|---|---|
| AdM | 21.228 | 17.328 | 3.545 |
| SNAIVE | 25.526 | 19.870 | 4.050 |
| AM | 54.187 | 48.292 | 9.845 |

**Answer:** The test-set RMSE values are given in the table. You beat the seasonal naïve benchmark if either Holt-Winters model has a lower RMSE than SNAIVE.

## 9 Exercise 9. STL + Box-Cox + ETS on the same retail series

```
lambda_retail <- retail_train %>% features(Turnover, guerrero) %>%
↪  pull(lambda_guerrero)

retail_stl_fit <- retail_train %>%
  model(
    STL_ETS = decomposition_model(
      STL(box_cox(Turnover, lambda_retail)),
      ETS(season_adjust ~ error("A") + trend("A") + season("N"))
    ),
    Best_HW = ETS(Turnover ~ error("A") + trend("A") + season("M"))
  )

retail_stl_fc <- forecast(retail_stl_fit, new_data = retail_test)
accuracy(retail_stl_fc, retail_test) %>%
```

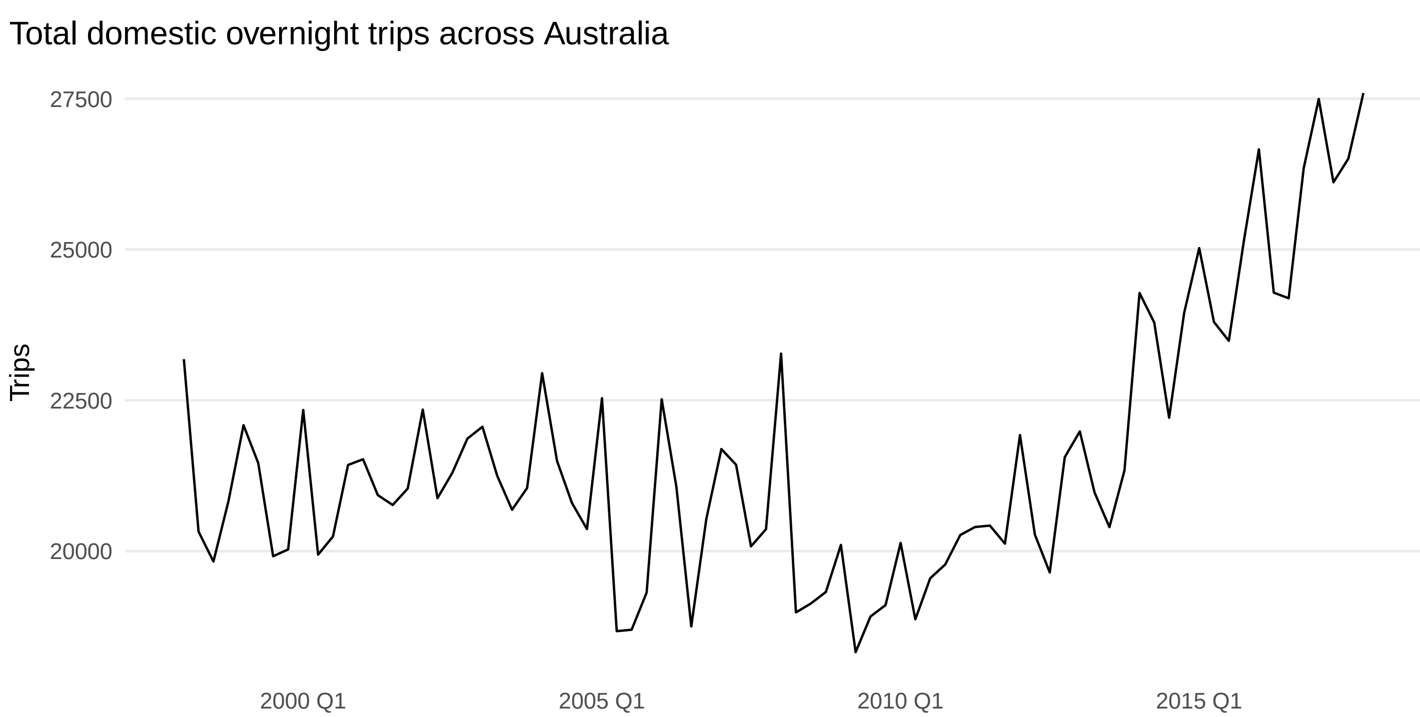
```r
  select(.model, RMSE, MAE, MAPE) %>%
  arrange(RMSE) %>%
  kable(digits = 3)
```

| .model | RMSE | MAE | MAPE |
|---|---|---|---|
| STL_ETS | 20.553 | 16.629 | 3.445 |
| Best_HW | 54.592 | 48.724 | 9.935 |

**Answer:** The STL + Box-Cox + ETS approach is better on the test set if `STL_ETS` has a lower RMSE than `Best_HW`. The comparison table gives the direct answer.

## 10 Exercise 10. Total domestic overnight trips across Australia

```r
aus_trips <- tourism %>%
  index_by(Quarter) %>%
  summarise(Trips = sum(Trips), .groups = "drop")

autoplot(aus_trips, Trips) +
  labs(title = "Total domestic overnight trips across Australia", y = "Trips")
```




### 10.1 Exercise 10(a) Plot the data and describe the main features.

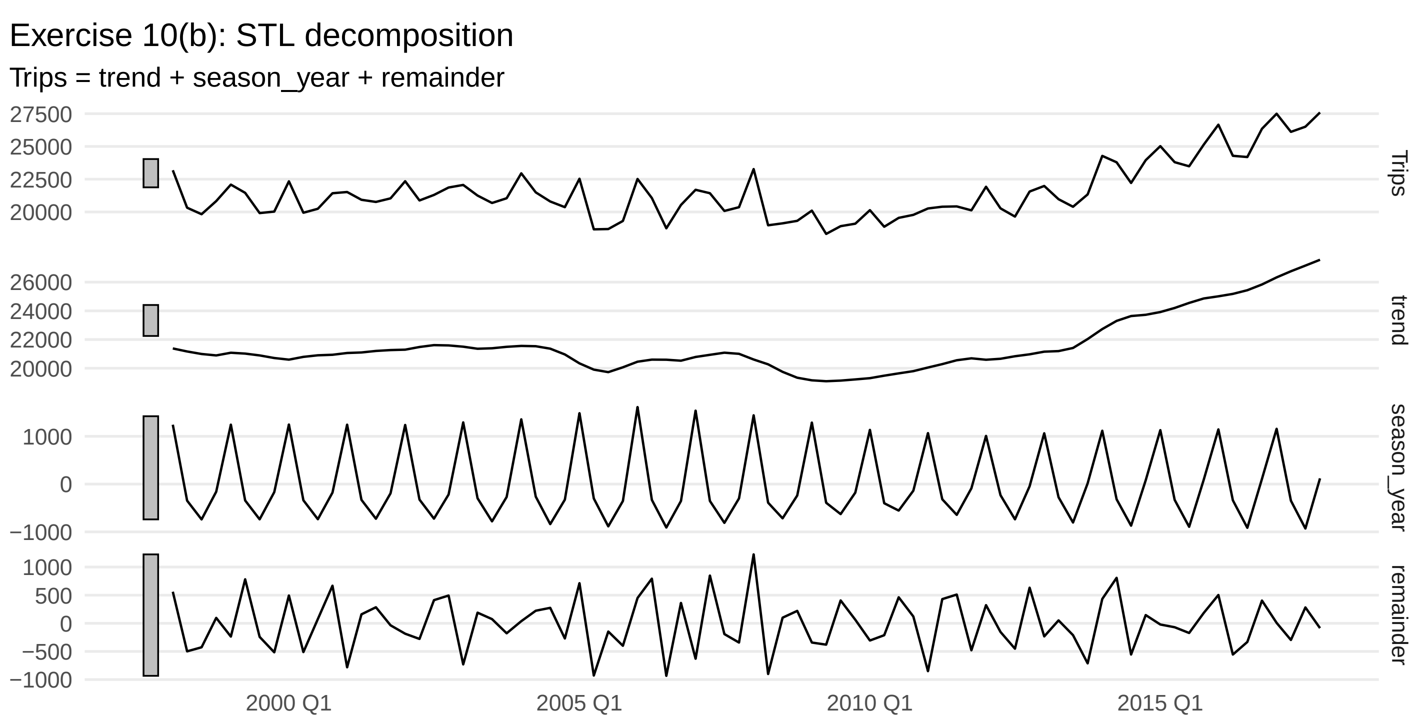
**Answer:** The series is quarterly and strongly seasonal. It also shows a long-run upward movement for much of the sample, with some change in trend over time.

## 10.2 Exercise 10(b) Decompose using STL and obtain the seasonally adjusted series.

```
aus_trips_stl <- aus_trips %>%
  model(STL(Trips)) %>%
  components()

aus_trips_stl %>%
  select(Quarter, Trips, season_adjust) %>%
  head() %>%
  kable(digits = 2)
```

| Quarter | Trips | season_adjust |
|---|---|---|
| 1998 Q1 | 23182.20 | 21939.48 |
| 1998 Q2 | 20323.38 | 20669.79 |
| 1998 Q3 | 19826.64 | 20566.03 |
| 1998 Q4 | 20830.13 | 20989.29 |
| 1999 Q1 | 22087.35 | 20843.24 |
| 1999 Q2 | 21458.37 | 21801.33 |

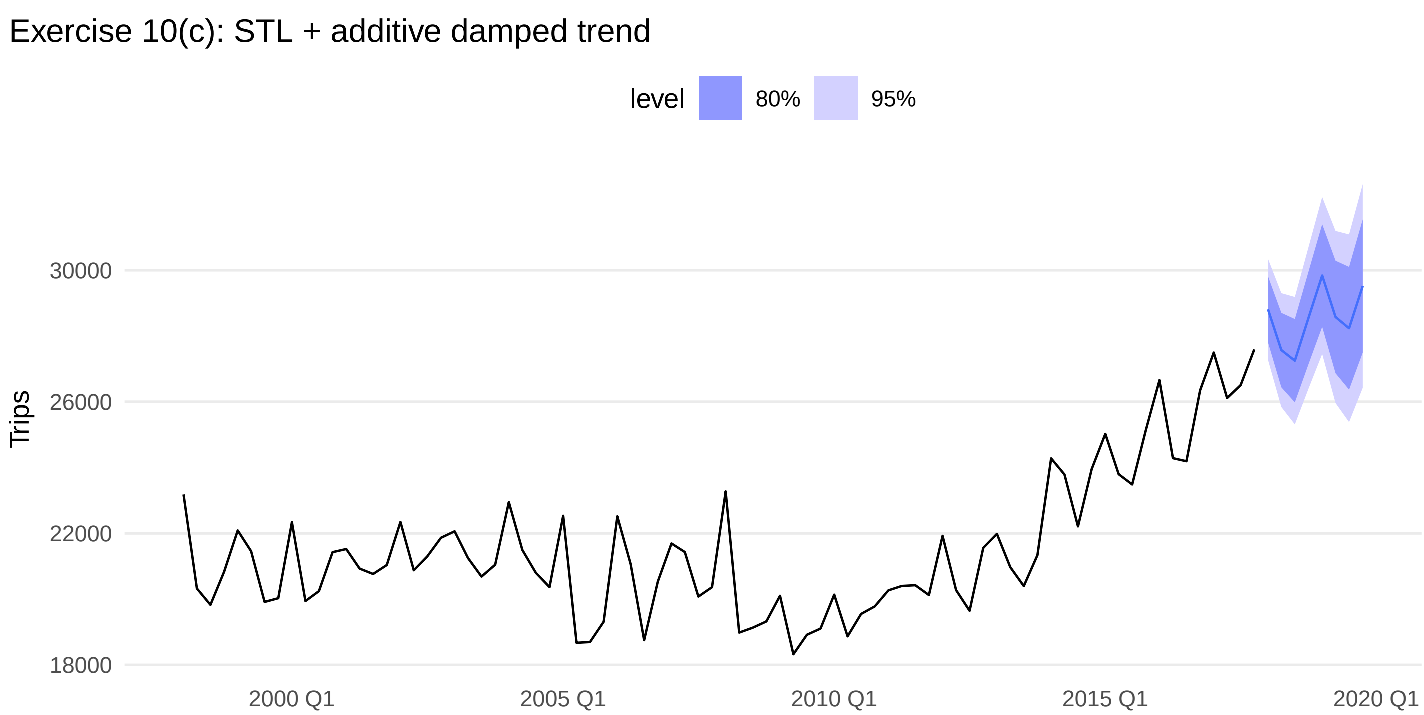
```
autoplot(aus_trips_stl) +
  labs(title = "Exercise 10(b): STL decomposition")
```




**Answer:** The `season_adjust` column shown above is the seasonally adjusted series. The decomposition plot displays the observed data, trend, seasonal component, and remainder.

### 10.3 Exercise 10(c) Forecast two years using additive damped trend on the seasonally adjusted data.

```
aus_trips_dcmp_fit <- aus_trips %>%
  model(
    STL_Ad = decomposition_model(
      STL(Trips),
      ETS(season_adjust ~ error("A") + trend("Ad") + season("N"))
    )
  )

aus_trips_dcmp_fc_Ad <- forecast(aus_trips_dcmp_fit, h = "2 years")

autoplot(aus_trips_dcmp_fc_Ad, aus_trips) +
  labs(title = "Exercise 10(c): STL + additive damped trend", y = "Trips")
```

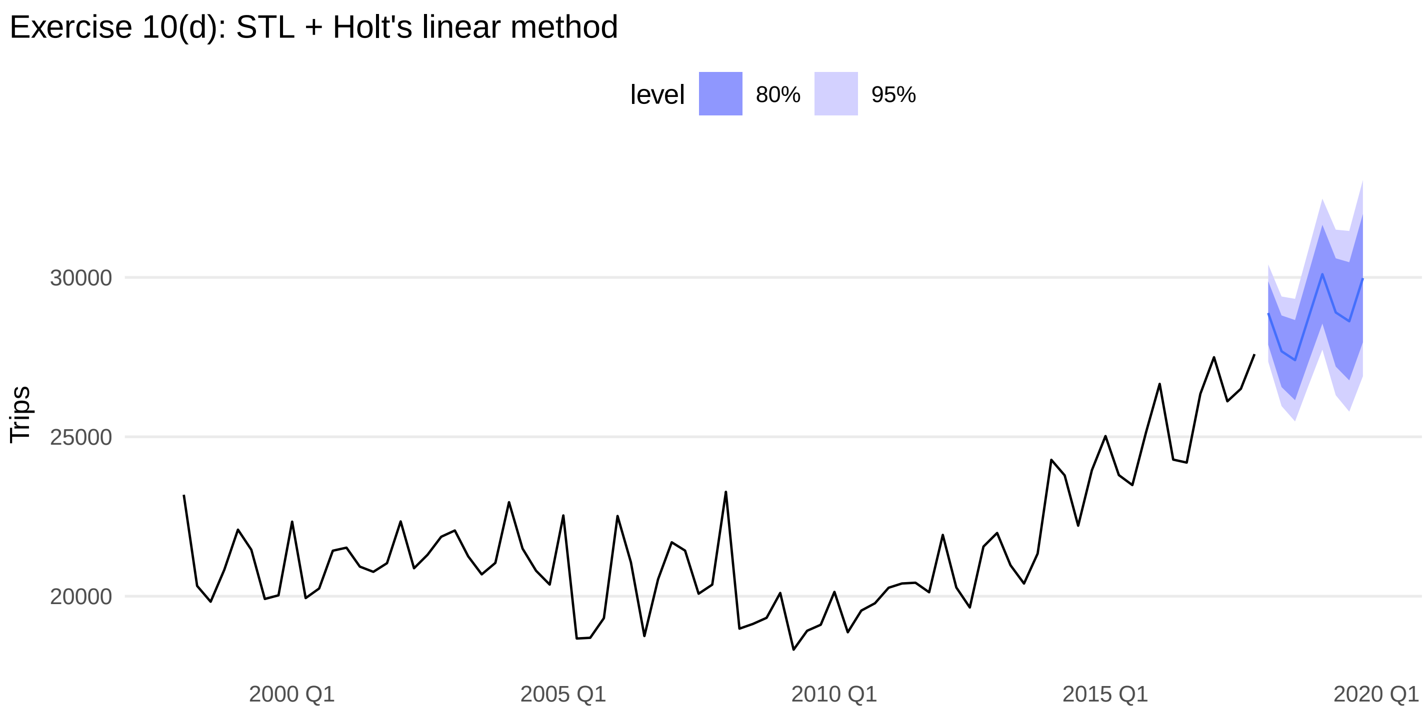



**Answer:** The required damped-trend forecast is the STL_Ad forecast shown above.

### 10.4 Exercise 10(d) Forecast two years using Holt's linear method on the seasonally adjusted data.

```
aus_trips_dcmp_fit2 <- aus_trips %>%
  model(
    STL_AA = decomposition_model(
      STL(Trips),
      ETS(season_adjust ~ error("A") + trend("A") + season("N"))
```

```
    )
  )

aus_trips_dcmp_fc_AA <- forecast(aus_trips_dcmp_fit2, h = "2 years")

autoplot(aus_trips_dcmp_fc_AA, aus_trips) +
  labs(title = "Exercise 10(d): STL + Holt's linear method", y = "Trips")
```

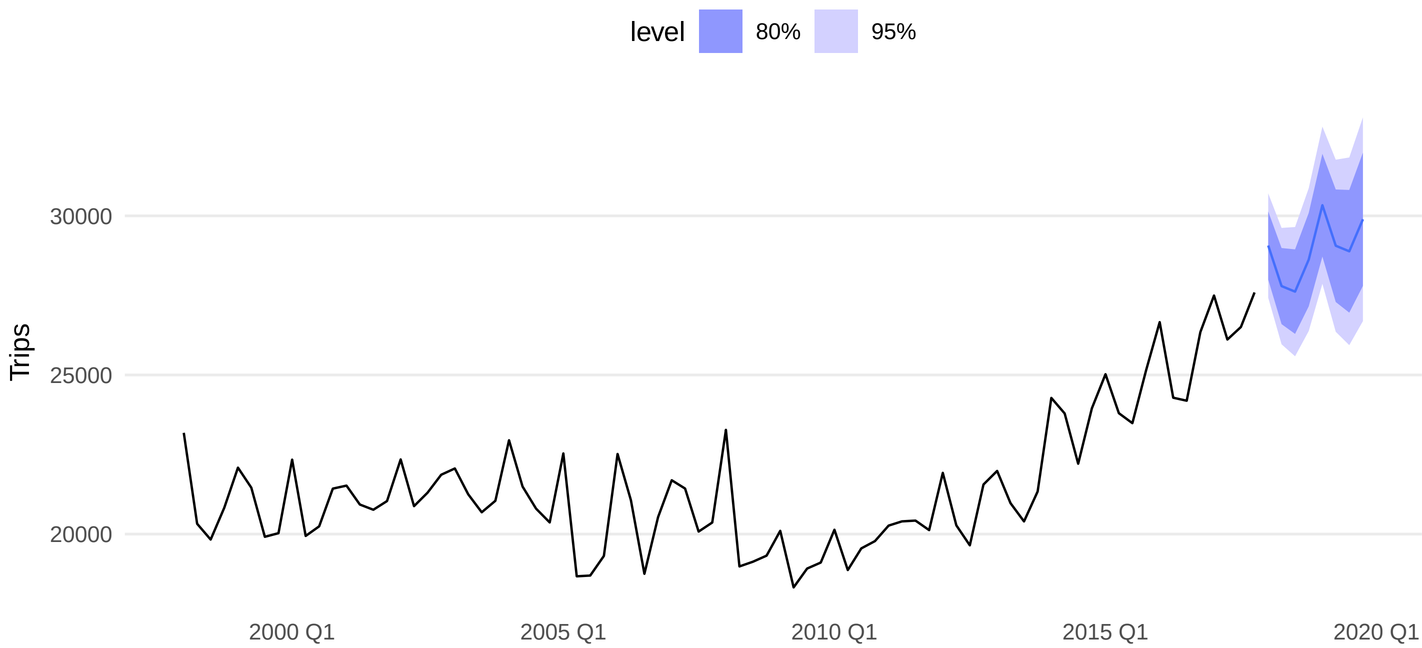



**Answer:** The required undamped Holt-style forecast is the STL_AA forecast shown above.

### 10.5 Exercise 10(e) Use ETS() to choose a seasonal model.

```
aus_trips_ets <- aus_trips %>% model(ETS = ETS(Trips))
aus_trips_ets_fc <- forecast(aus_trips_ets, h = "2 years")

autoplot(aus_trips_ets_fc, aus_trips) +
  labs(title = "Exercise 10(e): Automatically selected ETS model", y = "Trips")
```

Exercise 10(e): Automatically selected ETS model




**Answer:** The automatically selected seasonal ETS model is the one reported by `ETS()` for the ETS fit.

## 10.6 Exercise 10(f) Compare RMSE for the ETS model and the STL decomposition models.

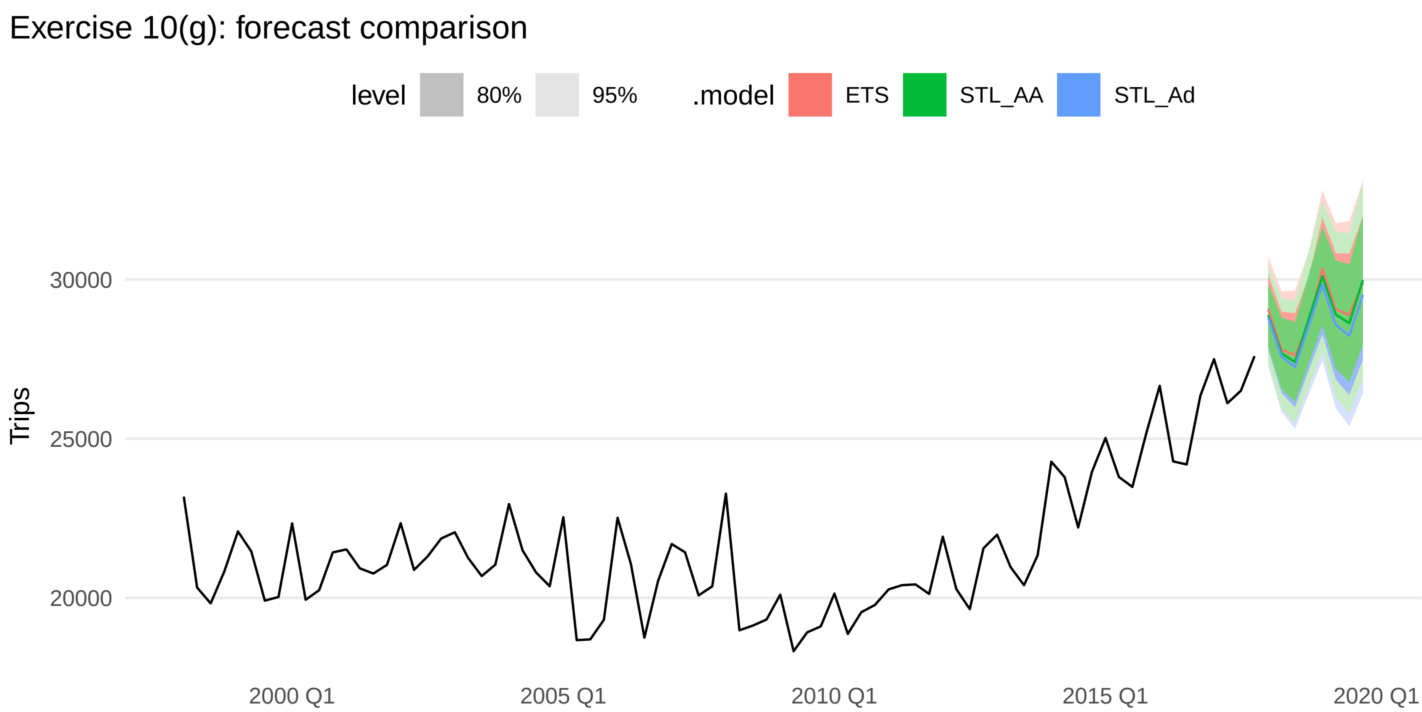
```r
trip_acc <- bind_rows(
  accuracy(aus_trips_dcmp_fit) %>% as_tibble(),
  accuracy(aus_trips_dcmp_fit2) %>% as_tibble(),
  accuracy(aus_trips_ets) %>% as_tibble()
)

trip_acc %>%
  select(.model, RMSE, MAE, MAPE) %>%
  arrange(RMSE) %>%
  kable(digits = 3)
```

| .model | RMSE | MAE | MAPE |
|---|---|---|---|
| STL_AA | 762.637 | 573.717 | 2.714 |
| STL_Ad | 763.369 | 576.166 | 2.722 |
| ETS | 793.670 | 604.109 | 2.857 |

**Answer:** The model with the smallest RMSE in the table has the best in-sample fit.

## 10.7 Exercise 10(g) Compare the forecasts. Which seems most reasonable?

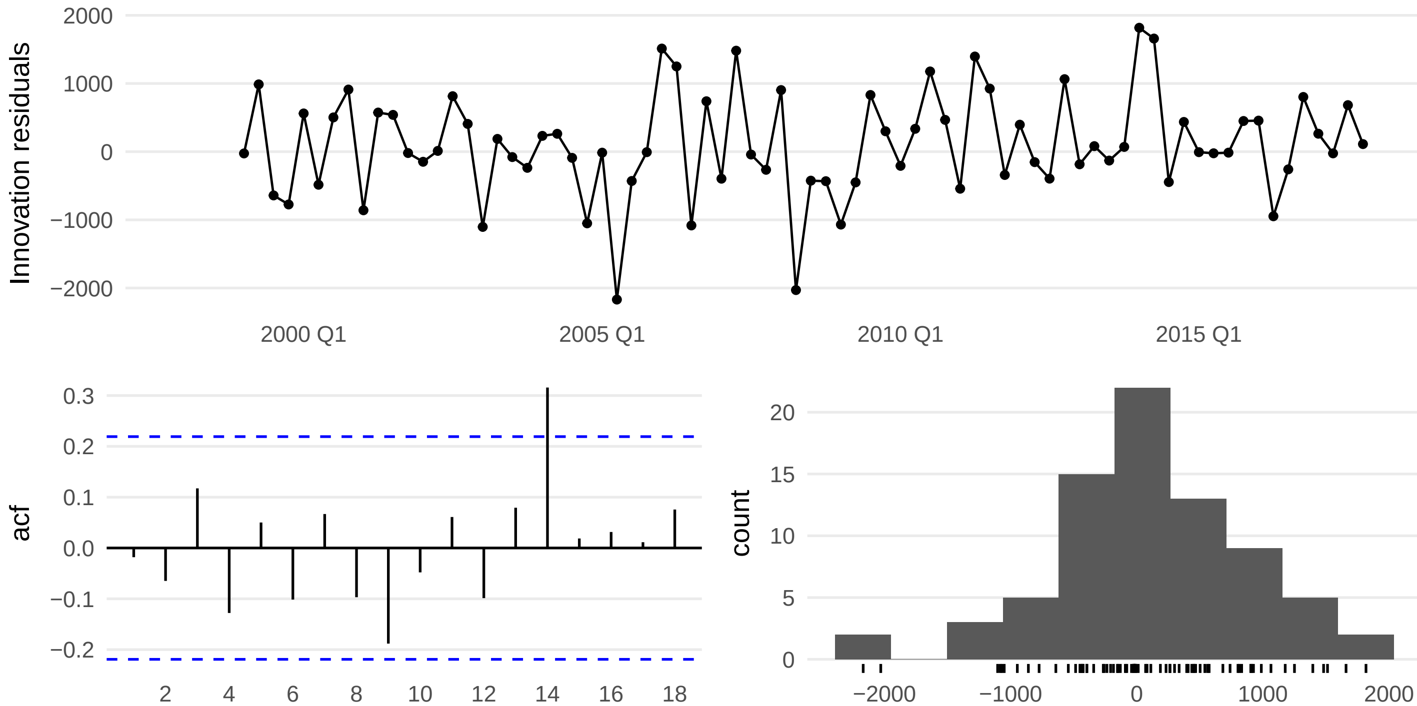
```r
bind_rows(aus_trips_dcmp_fc_Ad, aus_trips_dcmp_fc_AA, aus_trips_ets_fc) %>%
  autoplot(aus_trips) +
  labs(title = "Exercise 10(g): forecast comparison", y = "Trips")
```




**Answer:** The most reasonable forecast is usually the one that keeps the strong quarterly seasonality while avoiding implausibly strong long-run growth. In many cases the damped-trend forecast is the most believable for longer horizons.

### 10.8 Exercise 10(h) Check the residuals of the preferred model.

```r
best_trip_name <- trip_acc %>% arrange(RMSE) %>% slice(1) %>% pull(.model)
preferred_trip_fit <- switch(
  best_trip_name,
  STL_Ad = aus_trips_dcmp_fit %>% select(STL_Ad),
  STL_AA = aus_trips_dcmp_fit2 %>% select(STL_AA),
  ETS = aus_trips_ets %>% select(ETS)
)

gg_tsresiduals(preferred_trip_fit)
```



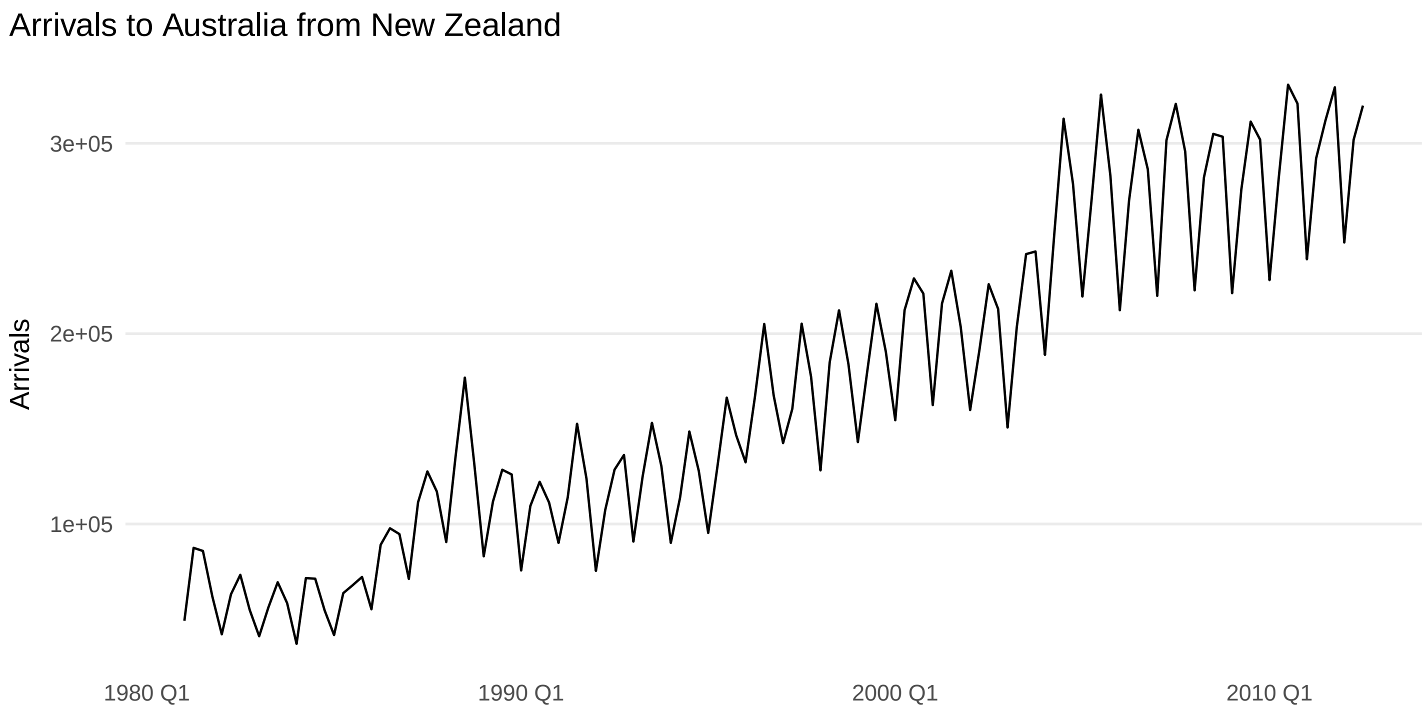

```
augment(preferred_trip_fit) %>% lb_tbl(lag = 8, dof = 0) %>% kable(digits = 4)
```

| .model | lb_stat | lb_pvalue |
|---|---|---|
| STL_AA | 5.1091 | 0.7458 |

**Answer:** The preferred model passes the residual check if the residuals show no obvious pattern and the Ljung-Box test does not reject white noise.

## 11 Exercise 11. Arrivals from New Zealand

```
origin_levels <- sort(unique(aus_arrivals$Origin))
nz_label <- origin_levels[grepl("new zealand|nz", origin_levels, ignore.case =
 ↪  TRUE)][1]

nz_arrivals <- aus_arrivals %>%
  filter(Origin == nz_label) %>%
  select(Quarter, Arrivals)

split_qtr <- yearquarter("2010 Q3")
nz_train <- nz_arrivals %>% filter(Quarter <= split_qtr)
nz_test <- nz_arrivals %>% filter(Quarter > split_qtr)

autoplot(nz_arrivals, Arrivals) +
  labs(title = "Arrivals to Australia from New Zealand", y = "Arrivals")
```




### 11.1 Exercise 11(a) Make a time plot and describe the main features.

**Answer:** The series is quarterly, strongly seasonal, and has a changing long-run level. Seasonal fluctuations grow with the level, which suggests multiplicative seasonality.

### 11.2 Exercise 11(b) Hold out the last two years and forecast using Holt-Winters multiplicative.

```
nz_hw <- nz_train %>%
  model(HW_Mult = ETS(Arrivals ~ error("A") + trend("A") + season("M")))

nz_hw_fc <- forecast(nz_hw, new_data = nz_test)
accuracy(nz_hw_fc, nz_test) %>%
  select(.model, RMSE, MAE, MAPE) %>%
  kable(digits = 3)
```

| .model | RMSE | MAE | MAPE |
|---|---|---|---|
| HW_Mult | 13720.4 | 11013.36 | 3.673 |

**Answer:** The two-year test-set accuracy for Holt-Winters multiplicative is given in the table above.

### 11.3 Exercise 11(c) Why is multiplicative seasonality necessary?

**Answer:** Multiplicative seasonality is necessary because the size of seasonal fluctuations increases as the level increases. The seasonal effect is proportional to the level rather than constant in absolute terms.

## 11.4 Exercise 11(d) Forecast the two-year test set using four methods.

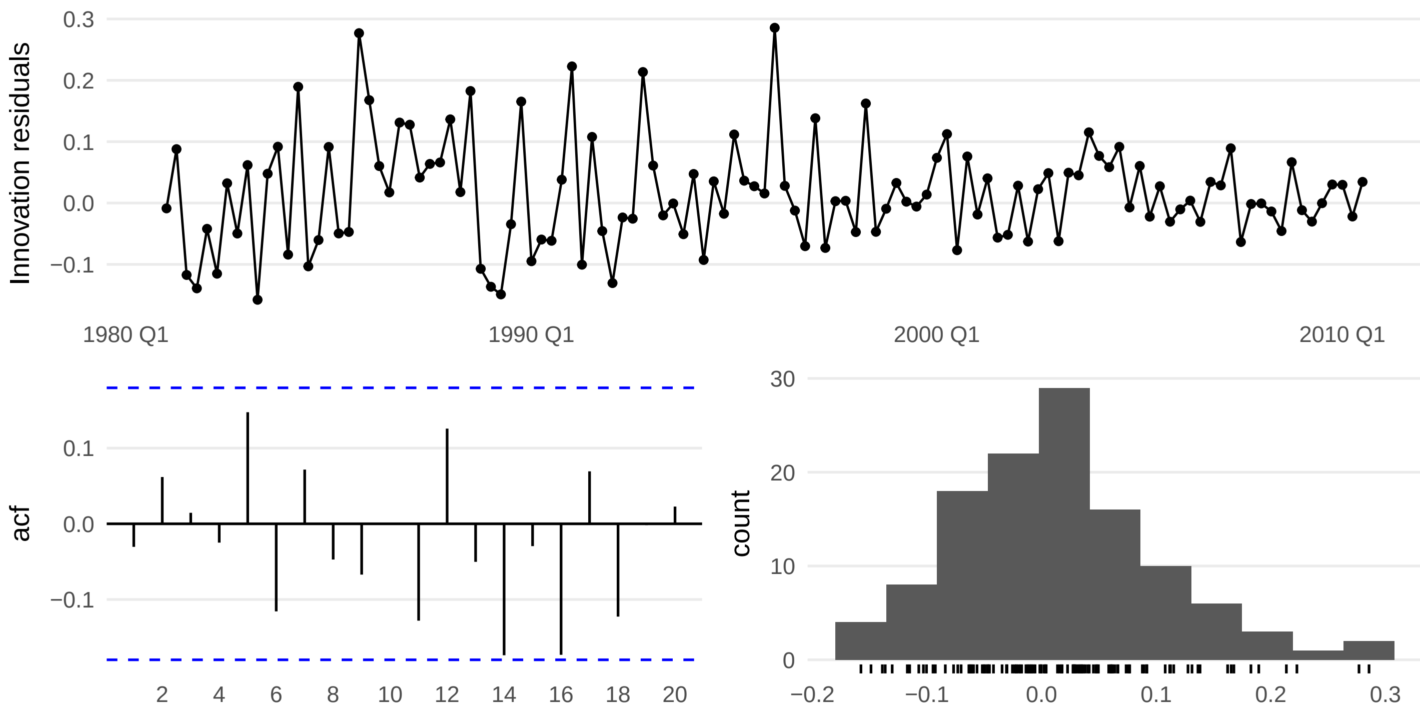
```r
nz_models <- nz_train %>%
  model(
    ETS_auto = ETS(Arrivals),
    Log_ETS = ETS(log(Arrivals)),
    SNAIVE = SNAIVE(Arrivals),
    STL_Log_ETS = decomposition_model(
      STL(log(Arrivals)),
      ETS(season_adjust ~ error("A") + trend("A") + season("N"))
    )
  )

nz_fc <- forecast(nz_models, new_data = nz_test)
accuracy(nz_fc, nz_test) %>%
  select(.model, RMSE, MAE, MAPE) %>%
  arrange(RMSE) %>%
  kable(digits = 3)
```

| .model | RMSE | MAE | MAPE |
|---|---|---|---|
| Log_ETS | 13342.04 | 11903.98 | 4.026 |
| ETS_auto | 14912.74 | 11420.69 | 3.783 |
| SNAIVE | 18051.08 | 17156.25 | 5.802 |
| STL_Log_ETS | 22722.78 | 16172.38 | 5.228 |

**Answer:** The table compares the four requested methods on the same two-year holdout sample.

## 11.5 Exercise 11(e) Which method is best? Does it pass the residual tests?

```r
nz_acc <- accuracy(nz_fc, nz_test)
best_nz_name <- nz_acc %>% arrange(RMSE) %>% slice(1) %>% pull(.model)
preferred_nz_fit <- nz_models %>% select(all_of(best_nz_name))

gg_tsresiduals(preferred_nz_fit)
```




```
augment(preferred_nz_fit) %>% lb_tbl(lag = 8, dof = 0) %>% kable(digits = 4)
```

| .model | lb_stat | lb_pvalue |
|---|---|---|
| Log_ETS | 6.0864 | 0.6376 |

**Answer:** The best method is the one with the lowest test-set RMSE in the table above. It passes the residual tests if the diagnostic plots do not show structure and the Ljung-Box p-value is not small.

### 11.6 Exercise 11(f) Compare the same four methods using time series cross-validation.

```
nz_cv <- nz_train %>%
  stretch_tsibble(.init = 5 * 4, .step = 1) %>%
  model(
    ETS_auto = ETS(Arrivals),
    Log_ETS = ETS(log(Arrivals)),
    SNAIVE = SNAIVE(Arrivals),
    STL_Log_ETS = decomposition_model(
      STL(log(Arrivals)),
      ETS(season_adjust ~ error("A") + trend("A") + season("N"))
    )
  ) %>%
  forecast(h = 8) %>%
  accuracy(nz_arrivals) %>%
  as_tibble()

nz_cv %>%
```

```r
  select(.model, RMSE, MAE, MAPE) %>%
  arrange(RMSE) %>%
  kable(digits = 3)
```

| .model | RMSE | MAE | MAPE |
|---|---|---|---|
| Log_ETS | 21357.56 | 15671.15 | 9.368 |
| ETS_auto | 22477.88 | 16330.39 | 9.681 |
| SNAIVE | 25530.02 | 18935.23 | 11.032 |
| STL_Log_ETS | 27039.91 | 19506.72 | 12.106 |

**Answer:** If the same method remains best under cross-validation, then the training/test result is reinforced. If a different method wins under cross-validation, the single holdout split was probably sensitive to the choice of cutoff.

# 12 Exercise 12. Cross-validation for Portland cement

## 12.1 Exercise 12(a) Produce 1-year-ahead ETS and seasonal naive forecasts using a stretching window.

## 12.2 Exercise 12(b) Compute the MSE of the 4-step-ahead errors and comment.

```r
cement <- aus_production %>% select(Quarter, Cement)

cement_cv <- cement %>%
  stretch_tsibble(.init = 5 * 4, .step = 1) %>%
  model(
    ETS = ETS(Cement),
    SNAIVE = SNAIVE(Cement)
  ) %>%
  forecast(h = 4)

cement_mse <- cement_cv %>%
  accuracy(cement, measures = list(MSE = MSE)) %>%
  as_tibble()

cement_mse %>%
  select(.model, MSE) %>%
  arrange(MSE) %>%
  kable(digits = 3)
```

| .model | MSE |
|---|---|
| ETS | 12613.48 |
| SNAIVE | 19373.14 |

**Answer:** The smaller 4-step-ahead MSE identifies the more accurate one-year-ahead forecasting method. Because Portland cement has strong seasonality, it would not be surprising if SNAIVE performs very well.

# 13 Exercise 13. Compare `ETS()`, `SNAIVE()`, and `decomposition_model(STL, ...)` on five series

```
compare_three_models <- function(data, value, test_years = 3) {
  value <- rlang::ensym(value)
  value_name <- rlang::as_name(value)
  idx <- tsibble::index_var(data)
  idx_name <- rlang::as_name(idx)

  if (nrow(data) == 0) {
    rlang::abort("compare_three_models() received an empty tsibble.")
  }

  n_per_year <- if (inherits(data[[idx_name]], "yearmonth")) {
    12
  } else if (inherits(data[[idx_name]], "yearquarter")) {
    4
  } else {
    1
  }

  test_n <- test_years * n_per_year

  if (nrow(data) <= test_n) {
    rlang::abort("compare_three_models() needs more observations than the requested
↪   test window.")
  }

  train <- head(data, nrow(data) - test_n)
  test <- tail(data, test_n)

  lambda_train <- train %>%
    features(!!value, guerrero) %>%
    pull(lambda_guerrero)

  train_bc <- train %>% mutate(value_bc = box_cox(!!value, lambda_train))
  test_bc <- test %>% mutate(value_bc = box_cox(!!value, lambda_train))

  fit_ets <- train %>% model(ETS = ETS(!!value))
  fit_snaive <- train %>% model(SNAIVE = SNAIVE(!!value))
  fit_stl <- train_bc %>%
    model(
      STL_ETS = decomposition_model(
        STL(value_bc),
        ETS(season_adjust)
      )
    )

  bind_rows(
```

```r
    forecast(fit_ets, new_data = test) %>% accuracy(test) %>% as_tibble(),
    forecast(fit_snaive, new_data = test) %>% accuracy(test) %>% as_tibble(),
    forecast(fit_stl, new_data = test_bc) %>% accuracy(test_bc) %>% as_tibble()
  ) %>%
    select(.model, RMSE, MAE, MAPE) %>%
    arrange(RMSE)
}
```

```r
beer <- aus_production %>% select(Quarter, Beer)
bricks <- aus_production %>% select(Quarter, Bricks)
a10 <- PBS %>% filter(ATC2 == "A10") %>% index_by(Month) %>% summarise(Cost =
 ↪  sum(Cost))
h02 <- PBS %>% filter(ATC2 == "H02") %>% index_by(Month) %>% summarise(Cost =
 ↪  sum(Cost))
food <- aus_retail %>%
  filter(Industry == "Food retailing") %>%
  index_by(Month) %>%
  summarise(Turnover = sum(Turnover))

res_beer <- compare_three_models(beer, Beer) %>% mutate(series = "Beer")
res_bricks <- compare_three_models(bricks, Bricks) %>% mutate(series = "Bricks")
res_a10 <- compare_three_models(a10, Cost) %>% mutate(series = "PBS A10")
res_h02 <- compare_three_models(h02, Cost) %>% mutate(series = "PBS H02")
res_food <- compare_three_models(food, Turnover) %>% mutate(series = "Food retailing")

ex13_results <- bind_rows(res_beer, res_bricks, res_a10, res_h02, res_food)

ex13_results %>%
  select(series, .model, RMSE, MAE, MAPE) %>%
  arrange(series, RMSE) %>%
  kable(digits = 3)
```

| series | .model | RMSE | MAE | MAPE |
|---|---|---:|---:|---:|
| Beer | STL_ETS | 0.078 | 0.070 | 0.608 |
| Beer | ETS | 9.621 | 8.919 | 2.133 |
| Beer | SNAIVE | 10.805 | 9.750 | 2.338 |
| Bricks | ETS | NaN | NaN | NaN |
| Bricks | SNAIVE | NaN | NaN | NaN |
| Bricks | STL_ETS | NaN | NaN | NaN |
| Food retailing | STL_ETS | 0.048 | 0.041 | 0.260 |
| Food retailing | ETS | 193.818 | 169.667 | 1.651 |
| Food retailing | SNAIVE | 699.075 | 624.750 | 5.856 |
| PBS A10 | STL_ETS | 4.158 | 3.550 | 1.962 |
| PBS A10 | ETS | 2360218.786 | 1851700.369 | 8.740 |
| PBS A10 | SNAIVE | 5180737.894 | 4330697.376 | 19.894 |
| PBS H02 | STL_ETS | 0.274 | 0.217 | 0.810 |
| PBS H02 | ETS | 76472.072 | 64455.509 | 7.046 |
| PBS H02 | SNAIVE | 85517.518 | 71610.554 | 7.885 |

**Answer:** For each of the five series, the best forecasting method is the method with the smallest test-set RMSE within that series. The table provides the full comparison for beer, bricks, diabetes drug subsidies, corticosteroid drug subsidies, and Australian food retailing turnover.

# 14 Exercise 14. Use ETS() to select models for several series

## 14.1 Exercise 14(a) Apply ETS() to total trips in tourism, the four gafa_stock series, and the lynx series in pelt. Does it always give good forecasts?

```
trips_total <- tourism %>%
  index_by(Quarter) %>%
  summarise(Trips = sum(Trips), .groups = "drop")

stocks <- gafa_stock %>%
  as_tsibble(index = Date, key = Symbol) %>%
  group_by(Symbol) %>%
  index_by(Week = yearweek(Date)) %>%
  summarise(Close = dplyr::last(Close))
lynx <- pelt %>% select(Year, Lynx)

trip_fit <- trips_total %>% model(ETS = ETS(Trips))
stock_fit <- stocks %>% model(ETS = ETS(Close))
lynx_fit <- lynx %>% model(ETS = ETS(Lynx))

report(trip_fit)
```

```
## Series: Trips
## Model: ETS(A,A,A)
##   Smoothing parameters:
##     alpha = 0.4495675
##     beta  = 0.04450178
##     gamma = 0.0001000075
##
##   Initial states:
##      l[0]      b[0]      s[0]     s[-1]     s[-2]    s[-3]
##  21689.64 -58.46946 -125.8548 -816.3416 -324.5553 1266.752
##
##   sigma^2:  699901.4
##
##      AIC     AICc      BIC
## 1436.829 1439.400 1458.267
```

```
report(stock_fit)
```

```
## # A tibble: 4 x 10
##   Symbol .model  sigma2 log_lik   AIC  AICc   BIC    MSE   AMSE    MAE
##   <chr>  <chr>    <dbl>   <dbl> <dbl> <dbl> <dbl>  <dbl>  <dbl>  <dbl>
```

```
## 1 AAPL   ETS    0.00122  -1116. 2238. 2238. 2248.   22.8   46.0 0.0259
## 2 AMZN   ETS    0.00164  -1604. 3218. 3218. 3236. 1446.  2634.  0.0295
## 3 FB     ETS    0.00137  -1103. 2216. 2216. 2234.   23.5   38.6 0.0261
## 4 GOOG   ETS    0.00113  -1577. 3160. 3160. 3170.  680.  1094.  0.0237
```

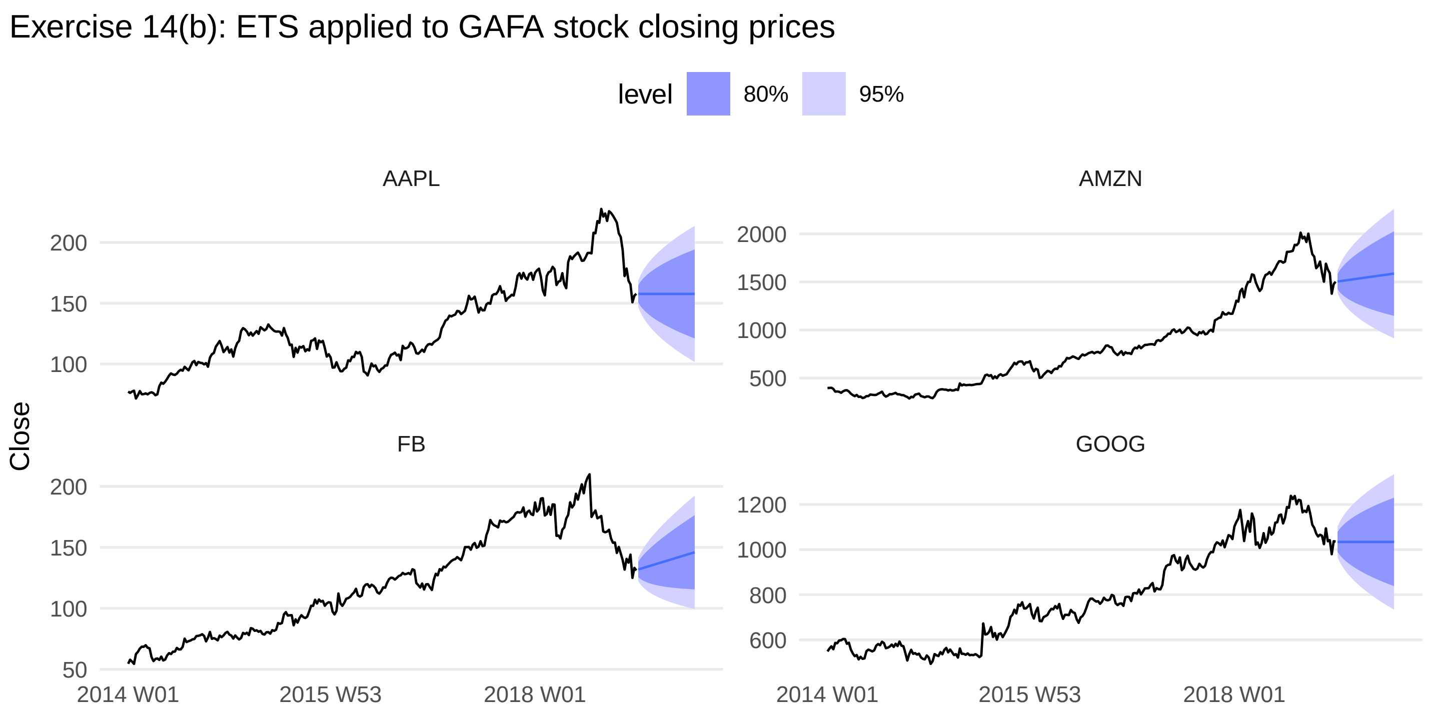
```r
report(lynx_fit)
```

```
## Series: Lynx
## Model: ETS(A,N,N)
##   Smoothing parameters:
##     alpha = 0.9998998
##
##   Initial states:
##      l[0]
##  25737.12
##
##   sigma^2:  162584145
##
##      AIC     AICc      BIC
## 2134.976 2135.252 2142.509
```

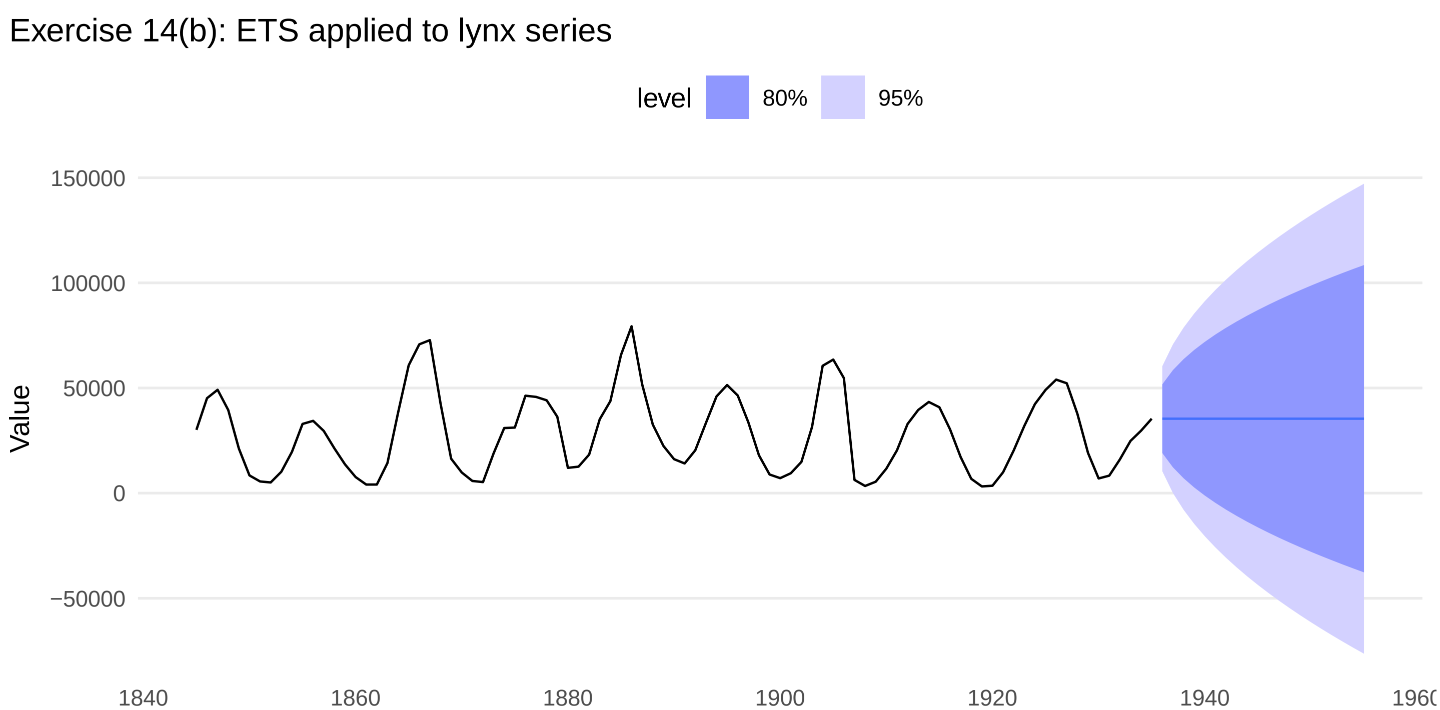
**Answer:** ETS() will always return a fitted exponential smoothing model, but that does **not** mean the forecasts are always good. It tends to work well for series with evolving level, trend, and seasonality, but it can struggle when the data behave more like random walks, structural breaks, bubbles, crashes, or cyclical ecological dynamics.

### 14.2 Exercise 14(b) Find an example where it does not work well. Why?

```r
stock_fc <- forecast(stock_fit, h = 30)
lynx_fc <- forecast(lynx_fit, h = 20)

autoplot(stock_fc, stocks) +
  facet_wrap(vars(Symbol), scales = "free_y") +
  labs(title = "Exercise 14(b): ETS applied to GAFA stock closing prices", y = "Close")
```




```r
autoplot(lynx_fc, lynx) +
  labs(title = "Exercise 14(b): ETS applied to lynx series", y = "Value")
```




**Answer:** A clear example is the stock-price series. Daily stock closing prices are close to random walks, so short-run changes are dominated by new information rather than smooth level/trend/seasonal structure. ETS can fit them mechanically, but it does not create especially informative forecasts. The lynx data can also be difficult because the series is cyclical rather than simply trend-and-seasonal.

## 15 Exercise 15. Show that ETS(M,A,M) point forecasts equal Holt-Winters multiplicative forecasts

**Answer:** Let the ETS(M,A,M) state-space equations be written in the same level, trend, and multiplicative seasonal components used by Holt-Winters multiplicative smoothing. The one-step conditional mean under ETS(M,A,M) is

$$\hat{y}_{t|t-1} = (\ell_{t-1} + b_{t-1})s_{t-m},$$

and the component updating equations are the same multiplicative Holt-Winters recursions:

$$\begin{aligned}
\ell_t &= \alpha \frac{y_t}{s_{t-m}} + (1-\alpha)(\ell_{t-1} + b_{t-1}), \\
b_t &= \beta^*(\ell_t - \ell_{t-1}) + (1-\beta^*)b_{t-1}, \\
s_t &= \gamma \frac{y_t}{\ell_{t-1} + b_{t-1}} + (1-\gamma)s_{t-m}.
\end{aligned}$$

Iterating these equations forward gives the $h$-step point forecast

$$\hat{y}_{t+h|t} = (\ell_t + hb_t)s_{t+h-m(k+1)},$$

which is exactly the Holt-Winters multiplicative forecast formula. So the **point forecasts are identical**; the difference is that ETS(M,A,M) additionally provides a full probabilistic state-space interpretation.

## 16 Exercise 16. Show that the forecast variance for ETS(A,N,N) is `sigma^2 [1 + alpha^2 (h-1)]`

**Answer:** For ETS(A,N,N), the observation and state equations are

$$y_t = \ell_{t-1} + \varepsilon_t, \qquad \ell_t = \ell_{t-1} + \alpha\varepsilon_t,$$

where the innovations $\varepsilon_t$ are independent with variance $\sigma^2$. The $h$-step-ahead forecast made at time $T$ is

$$\hat{y}_{T+h|T} = \ell_T.$$

The future observation can be expanded as

$$y_{T+h} = \ell_T + \alpha\varepsilon_{T+1} + \alpha\varepsilon_{T+2} + \cdots + \alpha\varepsilon_{T+h-1} + \varepsilon_{T+h}.$$

So the forecast error is

$$y_{T+h} - \hat{y}_{T+h|T} = \alpha\varepsilon_{T+1} + \cdots + \alpha\varepsilon_{T+h-1} + \varepsilon_{T+h}.$$

Because the innovations are independent,

$$\text{Var}(y_{T+h} - \hat{y}_{T+h|T}) = \alpha^2\sigma^2(h-1) + \sigma^2 = \sigma^2\left[1 + \alpha^2(h-1)\right].$$

Therefore the forecast variance is

$$\boxed{\sigma^2\left[1 + \alpha^2(h-1)\right].}$$

## 17 Exercise 17. Write down 95% prediction intervals for ETS(A,N,N)

**Answer:** From Exercise 16, the $h$-step forecast variance is

$$\sigma_h^2 = \sigma^2\left[1 + \alpha^2(h-1)\right].$$

For ETS(A,N,N), the point forecast at horizon $h$ is simply

$$\hat{y}_{T+h|T} = \ell_T.$$

Assuming normally distributed forecast errors, a 95% prediction interval is

$$\hat{y}_{T+h|T} \pm 1.96\sigma_h.$$

Substituting in the ETS(A,N,N) forecast and variance gives

$$\boxed{\ell_T \pm 1.96\,\sigma\sqrt{1 + \alpha^2(h-1)}}$$

or, written as lower and upper bounds,

$$\boxed{\left[\ell_T - 1.96\,\sigma\sqrt{1 + \alpha^2(h-1)},\ \ell_T + 1.96\,\sigma\sqrt{1 + \alpha^2(h-1)}\right].}$$